

Example 6.4: Aggregate surplus Consider a market with aggregate demand $x(p) = a - bp$ and aggregate supply curve $y(p) = J(p/2)$, where $a, b > 0$ and $J > 1$ denotes the number of firms in the industry. The CE price solves

$$a - bp = J \frac{p}{2}$$

or $p^* = 2a/(2b + J)$, which increases in the vertical intercept of aggregate demand but decreases in the number of firms. Therefore the equilibrium output is $x^* = a - b(2a/(2b + J)) = aJ/(2b + J)$. In this context, the surplus is

$$S(x^*) = \int_0^{x^*} p(x) - c'(x) dx,$$

where $p(x)$ is the inverse aggregate demand function. We solve for p in $x(p) = a - bp$ to obtain the indirect demand $p(x) = (a - x)/b$. Then, to find the aggregate marginal cost $c'(x)$, we solve for p in $y(p) = J(p/2)$ and get $p(x) = c'(x) = 2x/J$. Substituting these values yields

$$S(x^*) = \int_0^{aJ/(2b+J)} \left(\frac{a-x}{b} - \frac{2x}{J} \right) dx = \frac{a^2 J}{4b^2 + 2bJ},$$

which is increasing in the number of firms J , since $\partial S(x^*)/\partial J = a^2(2b + J)^2 > 0$. ■

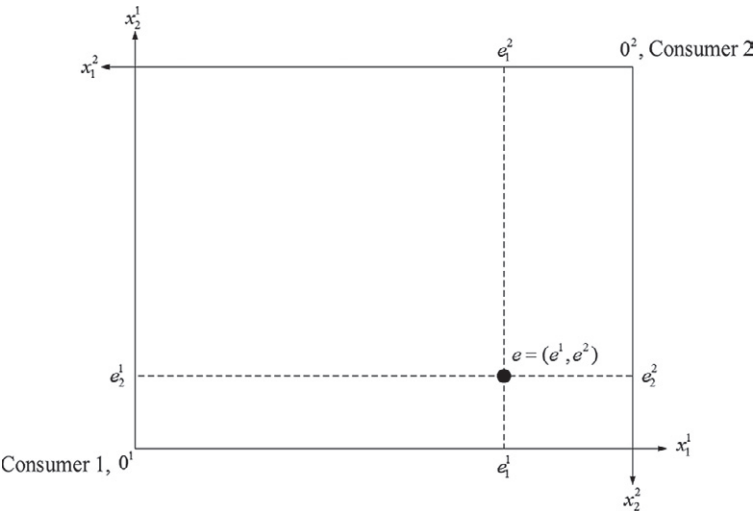
6.4 General Equilibrium

We now extend our discussion of equilibrium conditions from markets with a representative consumer to markets with multiple consumers (each consumer with potentially different preferences). We seek to evaluate under which price conditions the agents' demands for different goods are compatible with one another given the initial endowment of goods in the economy. For simplicity, we start with equilibrium allocations in economies without production (called “barter equilibrium,” since consumers exchange units of the goods they are initially endowed with), and subsequently analyze economies with production. At the end of the chapter we test our equilibrium results in large economies, and finally explore some comparative statics.

6.4.1 Economies without Production

Consider an economy with two goods and two consumers, $i = \{1, 2\}$, each initially endowed with $\mathbf{e}^i \equiv (e_1^i, e_2^i)$ units of good 1 and 2, respectively. Figure 6.12a depicts the

a.



b.

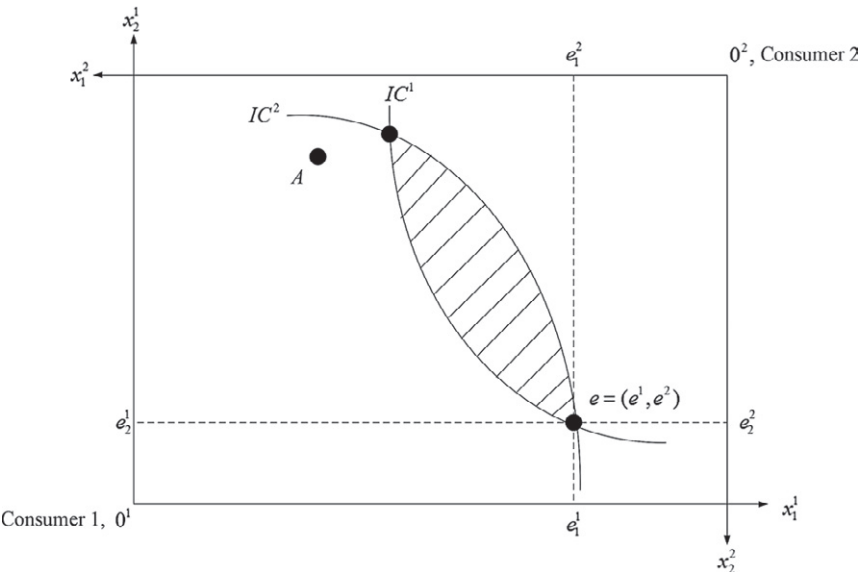


Figure 6.12
Two types of Edgeworth boxes

so-called Edgeworth box, with consumer 1's origin in the lower left-hand corner and consumer 2's origin at the opposite point of the box (the upper right-hand corner; if you cannot see that rotate the page 180 degrees). The figure also includes the initial endowment $\mathbf{e} \equiv (\mathbf{e}^1, \mathbf{e}^2)$, while any other allocation $\mathbf{x} \equiv (\mathbf{x}^1, \mathbf{x}^2)$ could similarly be depicted as a point in the box.

Figure 6.12b adds IC^1 , the indifference curve of consumer 1 passing through his endowment point $\mathbf{e}^1$, thus depicting bundles in the box that yield the same utility level as $\mathbf{e}^1$ for consumer 1. The figure also includes the indifference curve through the endowment point for consumer 2, IC^2 , and shades the region of bundles in the lens-shaped area between both consumers' indifference curves. More formally, the shaded area represents the set of bundles (x_1^1, x_2^1) for consumer 1 and (x_1^2, x_2^2) for consumer 2, satisfying

$$u^1(x_1^1, x_2^1) \geq u^1(e_1^1, e_2^1) \quad \text{and}$$

$$u^2(x_1^2, x_2^2) \geq u^2(e_1^2, e_2^2).$$

Hence a movement from the initial endowment $\mathbf{e}$ to allocation A (which lies outside the lens-shaped area) cannot be a barter equilibrium, since consumer 1 is worse off at A ; thus he would oppose a proposal to exchange $\mathbf{e}$ for A . Does that imply that any point in the lens-shaped area is a barter equilibrium? Not necessarily. Consider bundle B in figure 6.13. Despite lying inside the lens-shaped area, and thus yielding a higher utility level than the initial endowment $\mathbf{e}$ for both consumers, individuals could still find other points, such as D , that would make both of them better off than at B . Generally, any point on the CC curve depicted in figure 6.13 (often referred to as the “contract curve” in which indifference curves are tangent to one another) would be an equilibrium, since Pareto improvements are no longer possible. As we show in the next sections, while the contract curve depicts Pareto efficient allocations, only its portion lying inside the lens-shaped area constitutes a barter equilibrium.

The graphical presentation in figure 6.13 helps us in our initial search of a definition of equilibrium allocations. Nonetheless, before providing such a definition, we first need to define some additional ingredients. In particular, since allocations can only be part of an equilibrium if they are feasible, we still need to clarify which allocations are feasible, as well as which allocations can be blocked by one or more individuals in the economy.

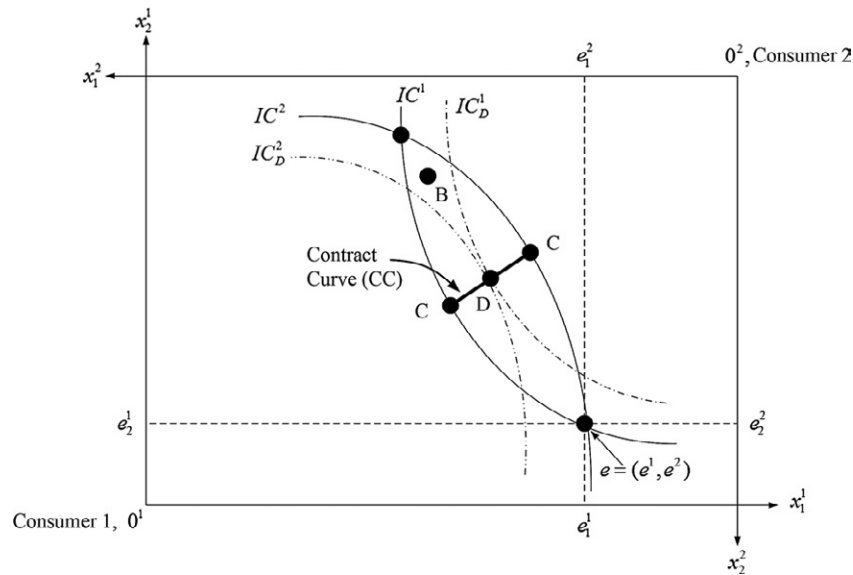


Figure 6.13
Contract curve

Feasible allocation An allocation $\mathbf{x} \equiv (\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^I)$ is feasible if it satisfies $\sum_{i=1}^I \mathbf{x}^i \leq \sum_{i=1}^I \mathbf{e}^i$.

That is, the aggregate amount of goods in allocation $\mathbf{x}$, when summing over all individuals $i = 1, 2, \dots, I$, does not exceed the aggregate initial endowment $\mathbf{e} \equiv \sum_{i=1}^I \mathbf{e}^i$.

Pareto efficient allocations (PEAs) A feasible allocation $\mathbf{x}$ is Pareto efficient if there is no other feasible allocation $\mathbf{y}$ that is weakly preferred by all consumers, meaning $\mathbf{y}^i \succeq \mathbf{x}^i$ for all $i \in I$; and is strictly preferred by at least one consumer i , $\mathbf{y}^i \succ \mathbf{x}^i$.

That is, allocation $\mathbf{x}$ is Pareto efficient if there is no other feasible allocation $\mathbf{y}$ making all individuals at least as well off as under $\mathbf{x}$ and making one or more individuals strictly better off. Intuitively, we cannot rearrange the bundles each consumer has in order to make at least one of them better off than under $\mathbf{x}$, without making others worse off. Mathematically, we can define the set of PEAs as the vector $(\mathbf{x}^1, \dots, \mathbf{x}^I)$ that solves

$$\max_{\mathbf{x}^1, \dots, \mathbf{x}^I \geq 0} u^1(\mathbf{x}^1)$$

subject to $u^j(\mathbf{x}^j) \geq \bar{u}^j$ for every individual $j \neq 1$, and

$$\sum_{i=1}^I \mathbf{x}^i \leq \sum_{i=1}^I \mathbf{e}^i \quad (\text{feasibility})$$

where $\mathbf{x}^i = (x_1^i, x_2^i)$. That is, an allocation $(\mathbf{x}^1, \dots, \mathbf{x}^I)$ is Pareto efficient if it maximizes individual 1's utility without reducing the utility of all other individuals below a given level $\bar{u}^j$ and satisfying feasibility (which in a two-consumer economy implies that $\mathbf{x}^1 + \mathbf{x}^2 \leq \mathbf{e}^1 + \mathbf{e}^2$).⁹ The Lagrangian associated with this maximization problem is

$$\begin{aligned} L(\mathbf{x}^1, \dots, \mathbf{x}^I; \lambda^2, \dots, \lambda^I, \mu) = & u^1(\mathbf{x}^1) + \lambda^2[u^2(\mathbf{x}^2) - \bar{u}^2] + \dots + \lambda^I[u^I(\mathbf{x}^I) - \bar{u}^I] \\ & + \mu \left[\sum_{i=1}^I \mathbf{e}^i - \sum_{i=1}^I \mathbf{x}^i \right]. \end{aligned}$$

Taking first-order conditions with respect to $\mathbf{x}^1 = (x_1^1, x_2^1)$ yields $\partial L / \partial x_k^1 = (\partial u^1(\mathbf{x}^1) / \partial x_k^1) - \mu \leq 0$ for every good $k = \{1, 2\}$ of consumer 1, whereas when we take first-order conditions with respect to $\mathbf{x}^j = (x_1^j, x_2^j)$ for every individual $j \neq 1$, we obtain $\partial L / \partial x_k^j = \lambda^j (\partial u^j(\mathbf{x}^j) / \partial x_k^j) - \mu \leq 0$. Finally, taking first-order conditions with respect to Lagrange multipliers λ^j and μ yields the constraints $u^j(\mathbf{x}^j) \geq \bar{u}^j$ and $\sum_{i=1}^I \mathbf{x}^i \leq \sum_{i=1}^I \mathbf{e}^i$, respectively. In the case of interior solutions, the combination of these first-order conditions produces a compact condition for Pareto efficiency

$$\frac{\partial u^1(\mathbf{x}^1) / \partial x_1^1}{\partial u^1(\mathbf{x}^1) / \partial x_2^1} = \frac{\partial u^j(\mathbf{x}^j) / \partial x_1^j}{\partial u^j(\mathbf{x}^j) / \partial x_2^j}, \quad \text{or} \quad MRS_{1,2}^1 = MRS_{1,2}^j$$

for every consumer $j \neq 1$. That is, the marginal rate of substitution between goods 1 and 2 ($MRS_{1,2}$) must coincide across all individuals in this economy.¹⁰ Graphically, their indifference curves become tangent to one another at the Pareto efficient allocations (PEAs). Intuitively, if we try to increase the utility of any consumer, we would need to make other consumer/s worse off. The next example applies this result to a setting where individual preferences are of the Cobb–Douglas type.

⁹ Generally, such a problem can be specified as maximizing the utility level of any individual i without reducing the utility level of any other individual j .

¹⁰ The result above easily extends to the case of economies with more than two goods, so that the MRS between any two goods k and l must coincide across all individuals in the economy, $MRS_{k,l}^1 = MRS_{k,l}^j$.

Example 6.5: Finding Pareto efficient allocations Consider a barter economy with two goods, 1 and 2, and two consumers, A and B , with initial endowments $\mathbf{e}^A = (100, 350)$ and $\mathbf{e}^B = (100, 50)$. For simplicity, assume both consumers' utility function is a Cobb–Douglas type given by $u^i(x_1^i, x_2^i) = x_1^i x_2^i$ for every individual $i = \{A, B\}$. Let us find the set of PEAs. Given regular preferences, such allocations are reached at points where the indifference curves of both consumers are tangent to one another, which is where their slopes satisfy $MRS_1^A = MRS_2^B$. In this context, $MRS_1^A = MRS_2^B$ implies $x_2^A/x_1^A = x_2^B/x_1^B$, or $x_2^A x_1^B = x_2^B x_1^A$. Using the feasibility requirement, $e_1^A + e_1^B = x_1^A + x_1^B$ for good 1 and $e_2^A + e_2^B = x_2^A + x_2^B$ for good 2, we obtain $x_1^B = e_1^A + e_1^B - x_1^A$ and $x_2^B = e_2^A + e_2^B - x_2^A$. Combining the tangency condition, $x_2^A x_1^B = x_2^B x_1^A$, and feasibility yields

$$x_2^A \underbrace{(e_1^A + e_1^B - x_1^A)}_{x_1^B} = \underbrace{(e_2^A + e_2^B - x_2^A)}_{x_2^B} x_1^A,$$

which can be rewritten as

$$x_2^A = \frac{e_2^A + e_2^B}{e_1^A + e_1^B} x_1^A = \frac{350 + 50}{100 + 100} x_1^A = \frac{400}{200} x_1^A,$$

or, more compactly, $x_2^A = 2x_1^A$ for all $x_1^A \in [0, 200]$. Figure 6.14 depicts the line representing the set of PEAs (the contract curve), $x_2^A = 2x_1^A$. ■

We are now ready to use these definitions in order to identify what we mean by an individual (or group of individuals) blocking a given allocation, that is, the formation of a blocking coalition of S individuals in an economy with I individuals.

Blocking coalitions Let $S \subset I$ denote a coalition of consumers. We say that S blocks the feasible allocation $\mathbf{x}$ if there is an allocation $\mathbf{y}$ meeting two conditions:

1. *Allocation $\mathbf{y}$ is feasible for S .* The aggregate amount of goods that individuals in S enjoy in allocation $\mathbf{y}$ coincides with their aggregate initial endowment, $\sum_{i \in S} \mathbf{y}^i = \sum_{i \in S} \mathbf{e}^i$.
2. *Allocation $\mathbf{y}$ is Pareto superior for S .* Allocation $\mathbf{y}$ makes all individuals in the coalition weakly better off than under $\mathbf{x}$, $\mathbf{y}^i \succeq \mathbf{x}^i$ where $i \in S$, and it makes at least one individual i strictly better off, $\mathbf{y}^i \succ \mathbf{x}^i$.

The following definitions form the “building blocks” of our definition of equilibrium in a barter economy:

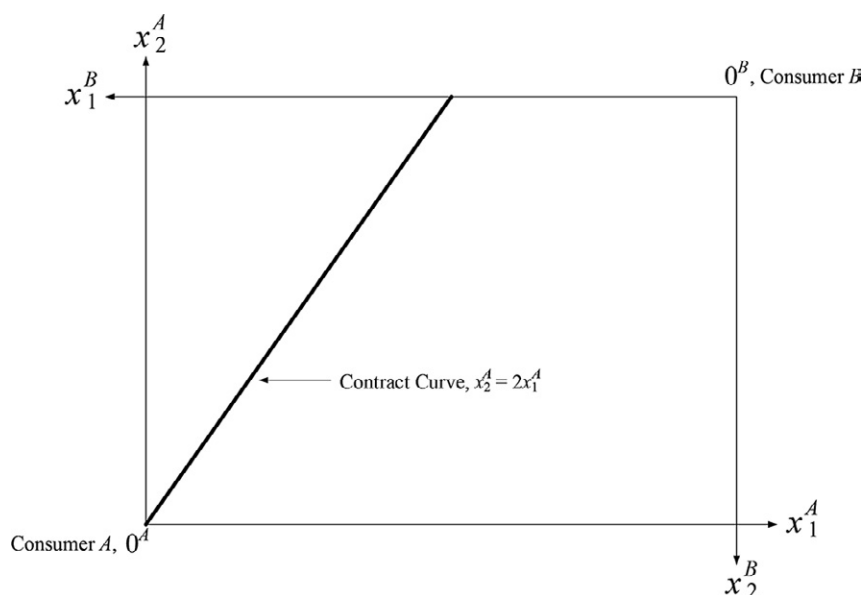


Figure 6.14
Contract curve $x_2^A = 2x_1^A$

Equilibrium A feasible allocation $\mathbf{x}$ is an equilibrium in the exchange economy with initial endowment $\mathbf{e}$ if $\mathbf{x}$ is *not blocked* by any coalition of consumers.

Intuitively, we can claim that a feasible allocation $\mathbf{x}$ is an equilibrium if there is no group of individuals S that could form a blocking coalition against $\mathbf{x}$ by finding a feasible allocation $\mathbf{y}$ that makes at least one of its members strictly better without harming any of the other members in S . Hence we can group together all equilibrium allocations in what is called the “core” of an exchange economy.

Core The core of an exchange economy with endowment $\mathbf{e}$, denoted $C(\mathbf{e})$, is the set of all unblocked feasible allocations (all equilibria).

Intuitively, it represents those allocations that are unblocked: (1) they are mutually beneficial for all individuals (i.e., they lie in the lens-shaped area); and (2) they do not allow for further Pareto improvements (i.e., lie on the contract curve). As depicted in figure 6.15, the set of core allocations is therefore the segment of the contract curve

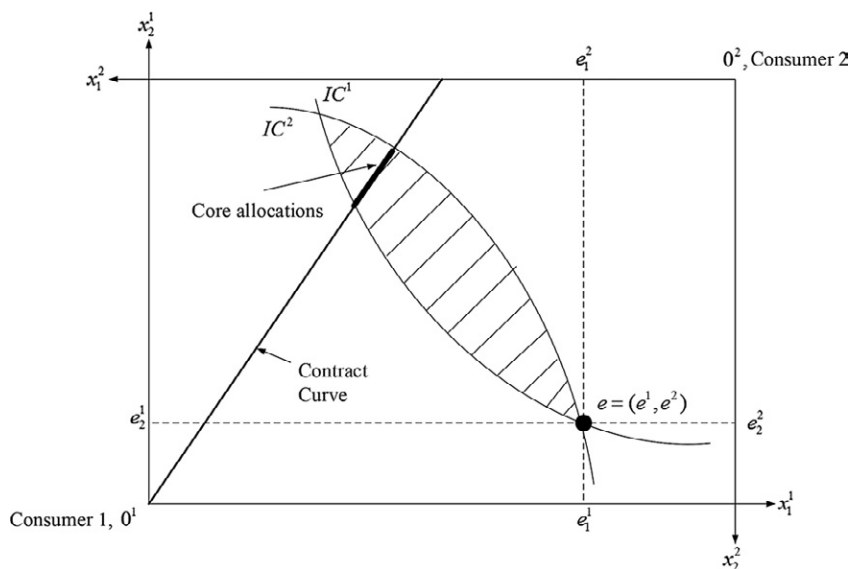


Figure 6.15
Core of an exchange economy

that lies within the lens-shaped area. Remarkably, these core allocations coincide with those in equilibria, as we show next.

6.4.2 Competitive Markets

In the previous barter economy we did not require prices. Let us now explore the notion of equilibrium in economies where we do allow prices to emerge. For presentation purposes, we first describe consumers' preferences, next the excess demand function that results from comparing the aggregate demand under a specific market price against the total endowment of each good. Then we define equilibrium allocations in competitive markets (which we refer as "Walrasian equilibrium allocations") and explore conditions under which they exist.

Consumers We consider consumers' utility functions to be continuous, strictly increasing, and strictly quasi-concave in $\mathbb{R}_+^N$ (recall that strictly quasi-concavity entails strictly convex indifference curves).

As a consequence the UMP of every consumer i , when facing a budget constraint

$$\mathbf{p} \cdot \mathbf{x}^i \leq \mathbf{p} \cdot \mathbf{e}^i \quad \text{for all price vectors } \mathbf{p} \gg \mathbf{0},$$

yields a unique solution, denoted as the Walrasian demand $\mathbf{x}(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i)$. In addition, $\mathbf{x}(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i)$ is continuous in the price vector $\mathbf{p}$. Intuitively, note that individual i 's income comes from selling his endowment $\mathbf{e}^i$ at market prices $\mathbf{p}$, earning $\mathbf{p} \cdot \mathbf{e}^i = p_1 e_1^i + \dots + p_k e_k^i$ dollars to be used in the purchase of allocation $\mathbf{x}^i$.

We can add the Walrasian demand $\mathbf{x}(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i)$ for good k of every individual in the economy, obtaining the aggregate demand for good k , and compare it against the aggregate endowment of that good, which yields the *excess demand of good k* , as follows:

$$z_k(\mathbf{p}) \equiv \sum_{i=1}^I x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) - \sum_{i=1}^I e_k^i, \quad \text{where } z_k(\mathbf{p}) \in \mathbb{R}.$$

Hence, when $z_k(\mathbf{p}) > 0$, the aggregate demand for good k exceeds its aggregate endowment, and we say that there is “excess demand” of good k ; in contrast, when $z_k(\mathbf{p}) < 0$, the opposite argument applies, and we say there is “excess supply” of good k . Figure 6.16 depicts the difference $\sum_{i=1}^I x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) - \sum_{i=1}^I e_k^i$ for a given good k in the left panel, and the resulting excess demand function $z_k(\mathbf{p})$ in the right panel.

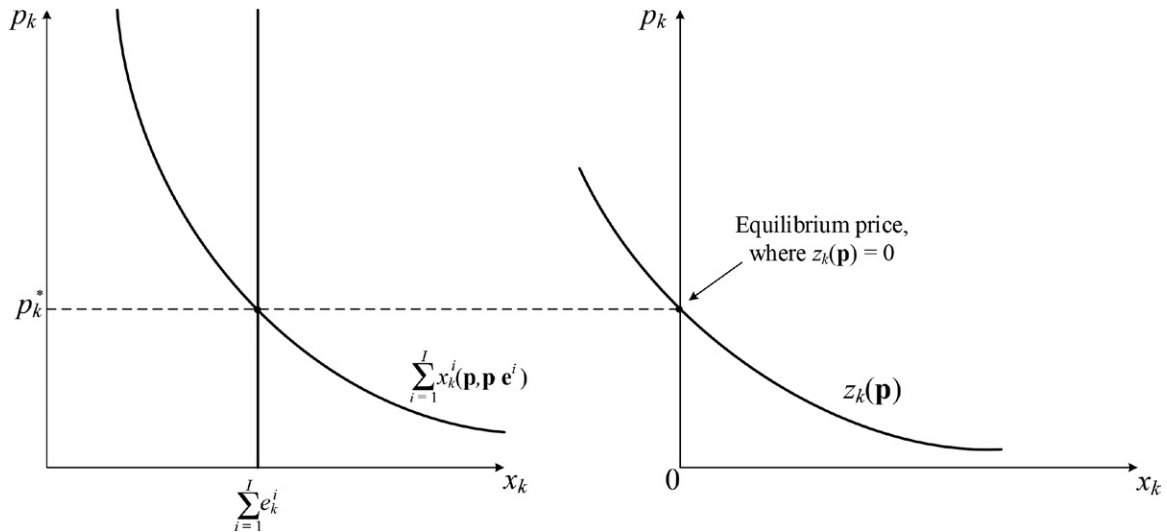


Figure 6.16
Difference in demand and supply, and excess demand

The excess demand function $\mathbf{Z}(\mathbf{p}) \equiv (z_1(\mathbf{p}), z_2(\mathbf{p}), \dots, z_L(\mathbf{p}))$ satisfies some interesting properties:

1. *Walras's law*, $\mathbf{p} \cdot \mathbf{z}(\mathbf{p}) = 0$ This follows from the property of a strictly increasing utility function: the budget constraint in the UMP binds for every consumer $i \in I$. In particular, since every consumer $i \in I$ exhausts all his income,

$$\sum_{k=1}^L p_k x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) = \sum_{k=1}^L p_k e_k^i \Leftrightarrow \sum_{k=1}^L p_k [x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) - e_k^i] = 0.$$

Summing over all individuals gives

$$\sum_{i=1}^I \sum_{k=1}^L p_k [x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) - e_k^i] = 0.$$

Since the order of summation is inconsequential, we can rewrite the expression above as

$$\sum_{k=1}^L \sum_{i=1}^I p_k [x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) - e_k^i] = 0,$$

which, in turn, is equivalent to

$$\sum_{k=1}^L p_k \underbrace{\left(\sum_{i=1}^I x_k^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) - \sum_{i=1}^I e_k^i \right)}_{z_k(\mathbf{p})} = 0$$

ultimately simplifying to

$$\sum_{k=1}^L p_k z_k(\mathbf{p}) = \mathbf{p} \cdot \mathbf{z}(\mathbf{p}) = 0.$$

In a two-good economy, Walras's law implies that $p_1 z_1(\mathbf{p}) = -p_2 z_2(\mathbf{p})$ indicating that, if there is excess demand in market 1, $z_1(\mathbf{p}) > 0$, then there must be excess supply in market 2, $z_2(\mathbf{p}) < 0$. Similarly, if market 1 is in equilibrium, $z_1(\mathbf{p}) = 0$, so is market 2, $z_2(\mathbf{p}) = 0$. More generally, if the markets of $L - 1$ goods are in equilibrium, so is the market for the L th commodity.

2. *Continuity*, $\mathbf{z}(\mathbf{p})$ is continuous in $\mathbf{p}$ This property follows from individual Walrasian demands being continuous in prices.
3. *Homogeneity of degree zero*, $z(\lambda \mathbf{p}) = z(\mathbf{p})$ for all $\lambda > 0$ This property follows from individual Walrasian demands being homogeneous of degree zero in prices, as described in previous chapters, which is to say, they are unaffected by an increase (or decrease) in all prices by a common factor $\lambda > 0$.

We can now use excess demand $\mathbf{z}(\mathbf{p})$ to define a Walrasian equilibrium allocation.

Walrasian equilibrium (WEA) A price vector $\mathbf{p}^* \gg 0$ is a Walrasian equilibrium if aggregate excess demand is zero at that price vector, $\mathbf{z}(\mathbf{p}^*) = 0$. In words, price vector $\mathbf{p}^*$ clears all markets.

Alternatively, $\mathbf{p}^* \gg 0$ is a Walrasian equilibrium if

1. each consumer solves his UMP, and
2. aggregate demand equals aggregate supply (i.e., markets clear), that is,

$$\sum_{i=1}^I x^i(\mathbf{p}, \mathbf{p} \cdot \mathbf{e}^i) = \sum_{i=1}^I \mathbf{e}^i.$$

Let us next explore existence and uniqueness of a Walrasian equilibrium.

Existence of WEA A Walrasian equilibrium price vector $\mathbf{p}^* \gg 0$, where $\mathbf{z}(\mathbf{p}^*) = 0$, exists if the excess demand function $\mathbf{z}(\mathbf{p})$ satisfies continuity and Walras's law (and both of these properties hold given the initial assumptions we imposed on utility functions).¹¹

Uniqueness of WEA

A desirable property of equilibrium prices is their uniqueness, as satisfied by the excess demand function in figure 6.17a but violated by that in figure 6.17b. As we next show, gross substitutability of all goods is a sufficient condition for an equilibrium price to be unique.

Proof By contradiction, suppose that there is another equilibrium price vector $\mathbf{p}'$, that is, $\mathbf{p}'$ satisfies $\mathbf{z}(\mathbf{p}') = 0$, where $\mathbf{p}', \mathbf{p}^* \gg 0$. Let us define price vector $\mathbf{p}'$ to be an increase in the price of all goods $j \neq k$ in vector $\mathbf{p}^*$ (except for the price of good k) as follows:

$$\mathbf{p}' = (mp_1^*, \dots, p_k^*, \dots, mp_L^*),$$

where $m > 1$. Hence, since the price of all other goods increased, the excess demand of good k must be positive; as prescribed by gross substitutability (p_j and x_k must move in the same direction, and then so do p_j and z_k). But then $z_k(\mathbf{p}') > 0$, implying that price vector $\mathbf{p}'$ does not clear the market for good k . Since all markets are not in equilibrium at $\mathbf{p}'$, price vector $\mathbf{p}'$ is not a WEA. (A similar argument applies if, rather than

¹¹ For a proof of that result, see Varian 1992, 321–22.

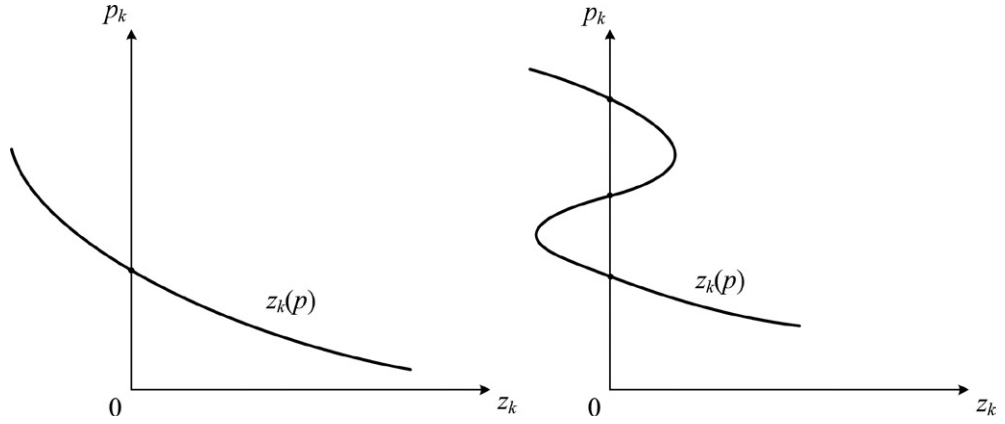


Figure 6.17
Unique and non-unique WEAs

increasing the price of all other goods, we decrease them. That is, if $0 < m < 1$, we obtain that the excess demand of good k satisfies $z_k(\mathbf{p}') < 0$.) ■

Example 6.6: Finding a Walrasian equilibrium allocation Continuing our example 6.5, we already determined that

$$MRS^A = MRS^B = \frac{p_1}{p_2},$$

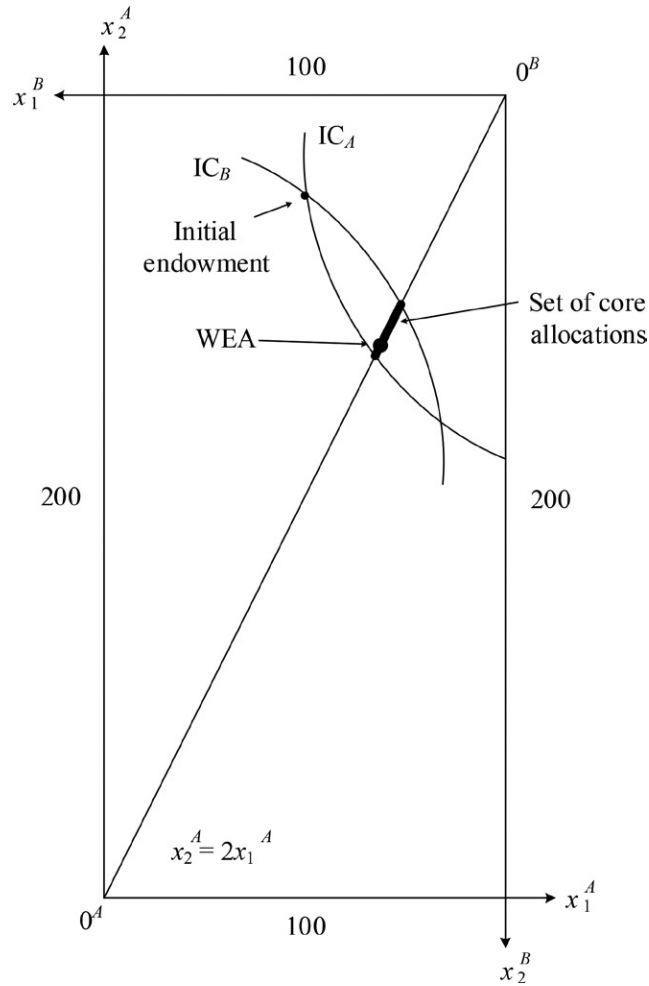
$$\frac{x_2^A}{x_1^A} = \frac{x_2^B}{x_1^B} = \frac{p_1}{p_2}.$$

From these two equations we can find the Walrasian demands for each good and each consumer. Starting with consumer A , we can rearrange the first and third terms of the preceding equation to obtain $p_1 x_1^A = p_2 x_2^A$. Inserting this into consumer A 's budget constraint yields

$$p_1 x_1^A + p_1 x_1^A = p_1(100) + p_2(350) \Rightarrow x_1^A = 50 + 175 \frac{p_2}{p_1},$$

which is consumer A 's demand for good 1. Plugging this value back into $p_1 x_1^A = p_2 x_2^A$ yields

$$p_1 \left(50 + 175 \frac{p_2}{p_1} \right) = p_2 x_2^A \Rightarrow x_2^A = 175 + 50 \frac{p_1}{p_2},$$

**Figure 6.18**

Initial allocation, core allocation, and WEA of example 6.6

which is consumer A 's demand for good 2. For consumer B , the process is similar except that we rearrange the second and third terms of our initial equation to obtain $p_1 x_1^B = p_2 x_2^B$. Inserting this into consumer B 's budget constraint yields

$$p_1 x_1^B + p_1 x_1^B = p_1(100) + p_2(50) \Rightarrow x_1^B = 50 + 25 \frac{p_2}{p_1},$$

which is consumer B 's demand for good 1. Substituting this value back into $p_1 x_1^B = p_2 x_2^B$ yields B 's demand for good 2

$$p_1 \left(50 + 25 \frac{p_2}{p_1} \right) = p_2 x_2^A \Rightarrow x_2^A = 25 + 50 \frac{p_1}{p_2}.$$

All that remains is to substitute each of our demands into our feasibility constraints and solve for relative prices. For good 1, the feasibility constraint is

$$x_1^A + x_1^B = 100 + 100,$$

$$50 + 175 \frac{p_2}{p_1} + 50 + 25 \frac{p_2}{p_1} = 100 + 200 \frac{p_2}{p_1} = 200.$$

which simplifies to $p_1/p_2 = 2$ (Note that the feasibility constraint for good 2 produces the same result.) Substituting the relative prices back into our Walrasian demands yields our Walrasian equilibrium, WEA,

$$\left(x_1^{A,*}, x_2^{A,*}; x_1^{B,*}, x_2^{B,*}; \frac{p_1}{p_2} \right) = (137.5, 275; 62.5, 125; 2). \blacksquare$$

Equilibrium Allocations Must Be in the Core

Our previous discussion suggested that Walrasian equilibrium allocations (WEAs) are mutually beneficial for all individuals. That is, an allocation cannot be blocked by any coalition of individuals, or, in other words, must be in the *core* of the economy. Let us next show that, if each consumer's utility function is strictly increasing, then every WEA is in the core, which we express as $W(\mathbf{e}) \subset C(\mathbf{e})$.

Proof Assume, by contradiction, a WEA, $\mathbf{x}(\mathbf{p}^*)$ with equilibrium price $\mathbf{p}^*$, that does not belong to the core, that is, $\mathbf{x}(\mathbf{p}^*) \notin C(\mathbf{e})$. Because $\mathbf{x}(\mathbf{p}^*)$ is a WEA, it must be feasible (as all equilibrium allocations must be feasible by definition). However, if such allocation is not part of the core, $\mathbf{x}(\mathbf{p}^*) \notin C(\mathbf{e})$, we can find a coalition of S individuals and another allocation $\mathbf{y}$ such that

$$u^i(\mathbf{y}^i) \geq u^i(\mathbf{x}^i(\mathbf{p}^*, \mathbf{p}^* \cdot \mathbf{e}^i)) \quad \text{for all } i \in S,$$

with strict inequality for at least one individual in the coalition, and that is feasible for the coalition, $\sum_{i \in S} \mathbf{y}^i = \sum_{i \in S} \mathbf{e}^i$. We can now multiply both sides of the feasibility condition by $\mathbf{p}^*$ to obtain

$$\mathbf{p}^* \sum_{i \in S} \mathbf{y}^i = \mathbf{p}^* \sum_{i \in S} \mathbf{e}^i.$$

However, if $\mathbf{x}^i(\mathbf{p}^*, \mathbf{p}^* \cdot \mathbf{e}^i)$ is a WEA, the preferable vector $\mathbf{y}^i$ must be more costly than $\mathbf{x}^i(\mathbf{p}^*, \mathbf{p}^* \cdot \mathbf{e}^i)$, that is,

$$\mathbf{p}^* \mathbf{y}^i \geq \mathbf{p}^* \mathbf{x}^i(\mathbf{p}^*, \mathbf{p}^* \cdot \mathbf{e}^i) = \mathbf{p}^* \mathbf{e}^i$$

with strict inequality for at least one individual. Summing over all consumers in the coalition S , we obtain

$$\mathbf{p}^* \sum_{i \in S} \mathbf{y}^i > \mathbf{p}^* \sum_{i \in S} \mathbf{x}^i(\mathbf{p}^*, \mathbf{p}^* \cdot \mathbf{e}^i) = \mathbf{p}^* \sum_{i \in S} \mathbf{e}^i,$$

thus contradicting $\mathbf{p}^* \sum_{i \in S} \mathbf{y}^i = \mathbf{p}^* \sum_{i \in S} \mathbf{e}^i$. Therefore all WEAs must be in the core, such that

$$\mathbf{x}(\mathbf{p}^*) \in C(\mathbf{e}). \blacksquare$$

We can immediately infer two conclusions from the previous result. On one hand, the core $C(\mathbf{e})$ will at least contain the WEA (or WEAs), implying that the core must be nonempty. On the other hand, since all core allocations are Pareto efficient, we cannot increase the welfare of one consumer without decreasing that of other consumers, implying that all WEAs (which are part of the core) are also Pareto efficient. This last result is often referred to as the *first welfare theorem*, as we compactly describe next.

First welfare theorem Every WEA is Pareto efficient.

As depicted in figure 6.19, the WEA lies on the core (the segment of the contract curve within the lens-shaped area), and the core is a subset of all PEAs (as illustrated in the contract curve). For a numerical example of the first welfare theorem, see example 6.6, where the WEA we found $(x_1^{A,*}, x_2^{A,*}; x_1^{B,*}, x_2^{B,*}; p_1/p_2) = (137.5, 275; 62.5, 125; 2)$, lies on the contract curve $x_2^A = 2x_1^A$, and hence it is Pareto efficient.

Consider the setting depicted in figure 6.20 and, starting from initial endowment $\mathbf{e}$, assume that the WEA is $\mathbf{x}'$, which also belongs to the set of core allocations $C(\mathbf{e})$. However, suppose that society would prefer allocations $\bar{\mathbf{x}}$ to $\mathbf{x}'$ according to some social welfare function that aggregates individual preferences over bundles.¹² A natural

¹² Most of the social welfare functions we describe aggregate individual utility functions, $W = f(u_1, \dots, u_I)$. Common examples are (1) the “utilitarian,” $W = \sum_{i=1}^I a_i u_i$, where $a_i \geq 0$ denotes the weight that society assigns to individual i ; (2) the Cobb–Douglas type, $W = \prod_{i=1}^I u_i^{b_i}$ or applying logs $\sum_{i=1}^I b_i \log u_i$, where $b_i \geq 0$; and (3) the “Benthamian” $W = \min\{u_1, \dots, u_I\}$, which is concerned about the welfare of the individual in the worst position of society.

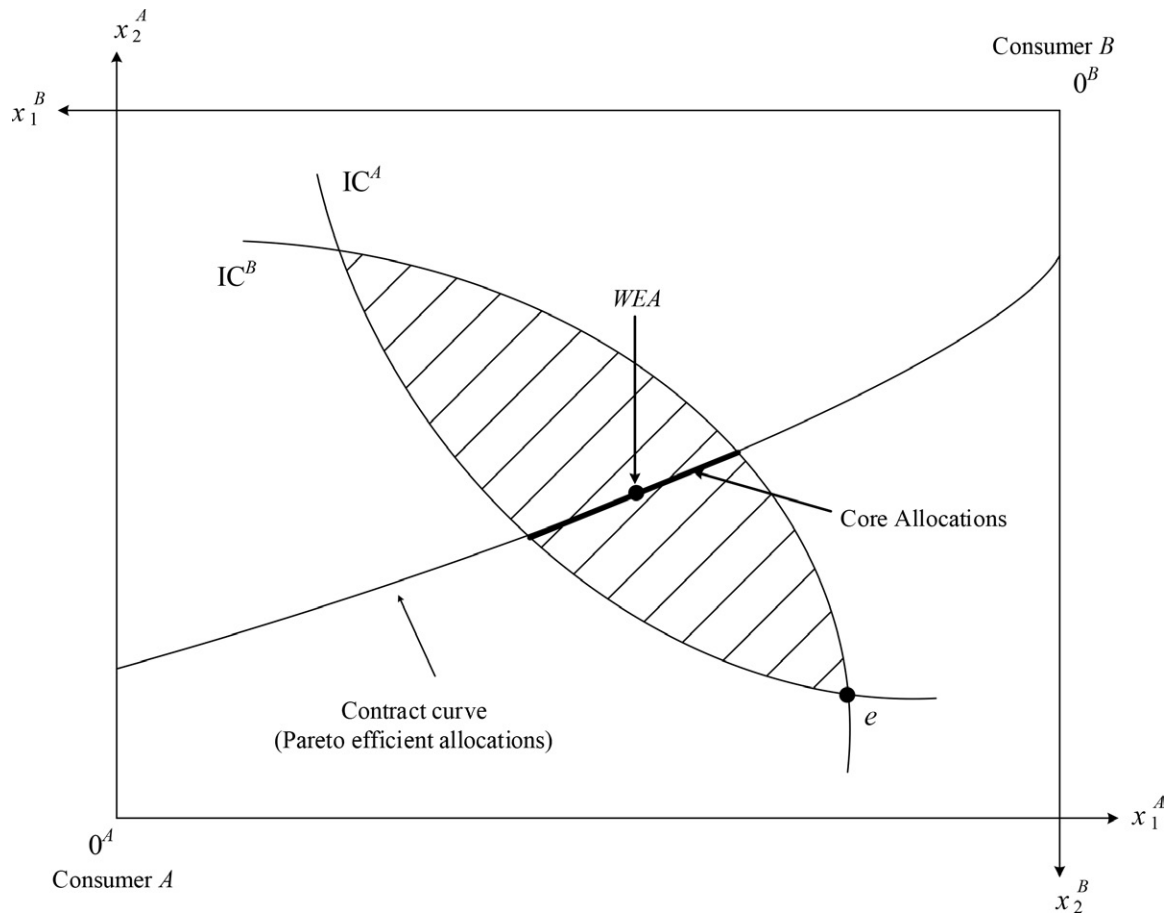


Figure 6.19
First welfare theorem

question is whether society could simply alter the initial endowment, from e to e'' (or generally, to any point e^{*i} on the budget line, satisfying $\mathbf{p}^* \cdot \mathbf{e}^{*i} = \mathbf{p}^* \cdot \bar{\mathbf{x}}^i$), and then “let the market work” (i.e., allow each individual consumer to independently solve his own UMP). Would that variation in the initial endowment (followed by letting the market work) lead to the desired WEA $\bar{\mathbf{x}}$? As shown by the second welfare theorem, the answer to this question is yes.

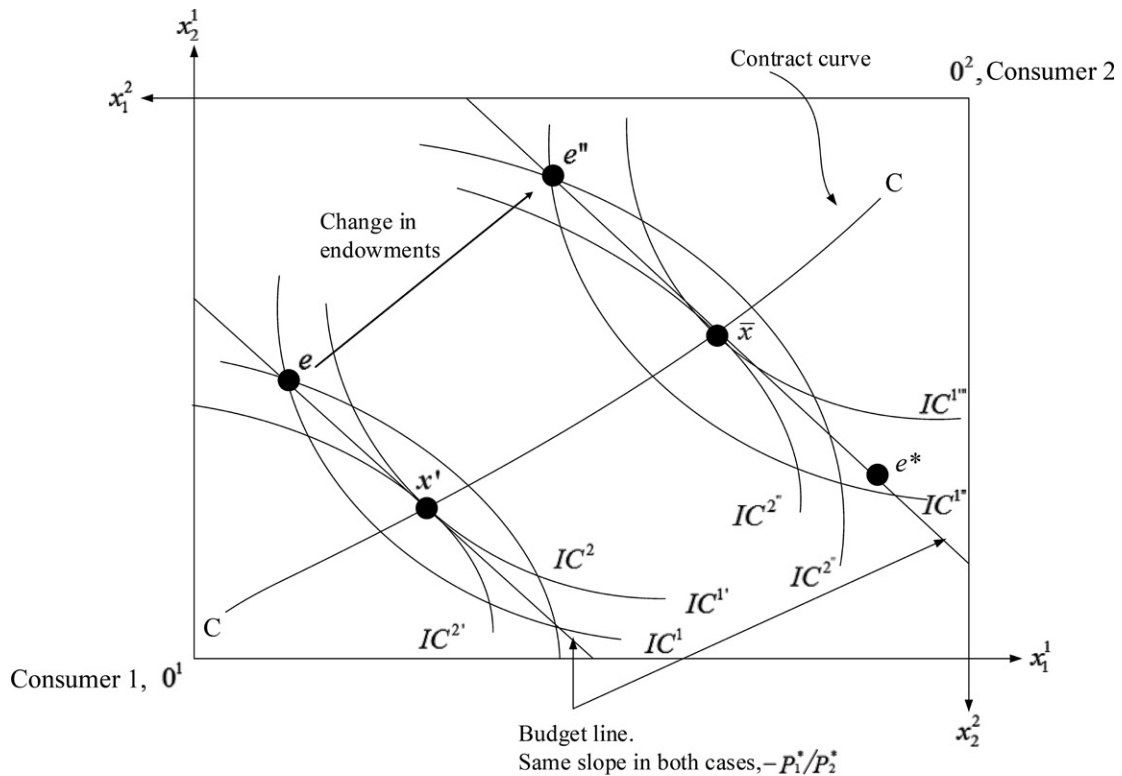


Figure 6.20
Second welfare theorem

Second welfare theorem A PEA $\bar{x}$ (i.e., $\bar{x}$ lies on the contract curve), and the endowments, are redistributed so that the new endowment vector e^{*i} lies on a line satisfying $\mathbf{p}^* \cdot e^{*i} = \mathbf{p}^* \cdot \bar{x}^i$ for every consumer i . Then the PEA $\bar{x}$ is a WEA given the new endowment vector e^* .

The first and second welfare theorems provide sharp results about the relationship between WEAs and PEAs when markets operate without distortions. However, when market failures exist—such as market power, externalities (in consumption or production), and public goods, or when some agents have access to information that other agents cannot accurately observe—these two theorems do not necessarily apply, as we examine in subsequent chapters. Likewise our previous analysis assumed that consumers have similar bargaining power when negotiating a price for each good. However,

in some settings one consumer can sustain all the bargaining power if, for instance, he makes a take-it-or-leave-it price offer to other consumers.¹³

Example 6.7: WEA and second welfare theorem Consider an economy with utility functions $u^A = x_1^A x_2^A$ for consumer A and $u^B = \min\{x_1^B, x_2^B\}$ for consumer B , where initial endowments are $e^A = (3, 1)$ and $e^B = (1, 3)$. First, let us find the set of PEAs (similarly as in example 6.5), then we will find the set of WEA, where we use good 2 as the numeraire, $p_2 = 1$. Finally, assuming that society seeks to implement allocation $\hat{x}^A = (1, 1)$ for consumer A and $\hat{x}^B = (3, 3)$ for consumer B , we find the initial endowments that achieve that this allocation becomes the WEA.

- *PEAs* Starting with consumer B , due to his preferences being perfect complements, consumer B wants to consume at the kink of his indifference curves, that is, $x_1^B = x_2^B$, by consuming goods 1 and 2 in equal quantities. From the feasibility constraints, we have

$$x_1^A + x_1^B = 4, \quad x_2^A + x_2^B = 4.$$

Then we can substitute x_2^B for x_1^B in the first feasibility condition, $x_1^B = x_2^B$, and solve for x_2^B , yielding $x_2^B = 4 - x_1^A$. Substituting this value into the second feasibility gives $x_2^A + \underbrace{(4 - x_1^A)}_{x_2^B} = 4 \Rightarrow x_2^A = x_1^A$, which defines our contract curve, the set of

PEAs, as depicted in figure 6.21.

- *WEA* Consumer A 's utility maximization problem is

$$\max_{x_1^A, x_2^A} x_1^A x_2^A$$

$$\text{subject to } p_1 x_1^A + x_2^A \leq p_1(3) + 1$$

we take first-order conditions to obtain

$$x_2^A - \lambda p_1 = 0$$

$$x_1^A - \lambda = 0,$$

¹³ For example, consumer A could announce a price ratio to individual B , who either accepts it or rejects it (which yields every consumer with his initial endowment). As in our previous discussion, consumer B would take the price ratio announced by A as given, and solve his UMP in order to find his offer curve. Consumer A , in contrast, would anticipate B 's offer curve, and use it as the constraint of his UMP to reach the highest possible utility level. As one can expect, the WEA that emerges in this context is not Pareto optimal, as we illustrate in exercise 14 at the end of the chapter.

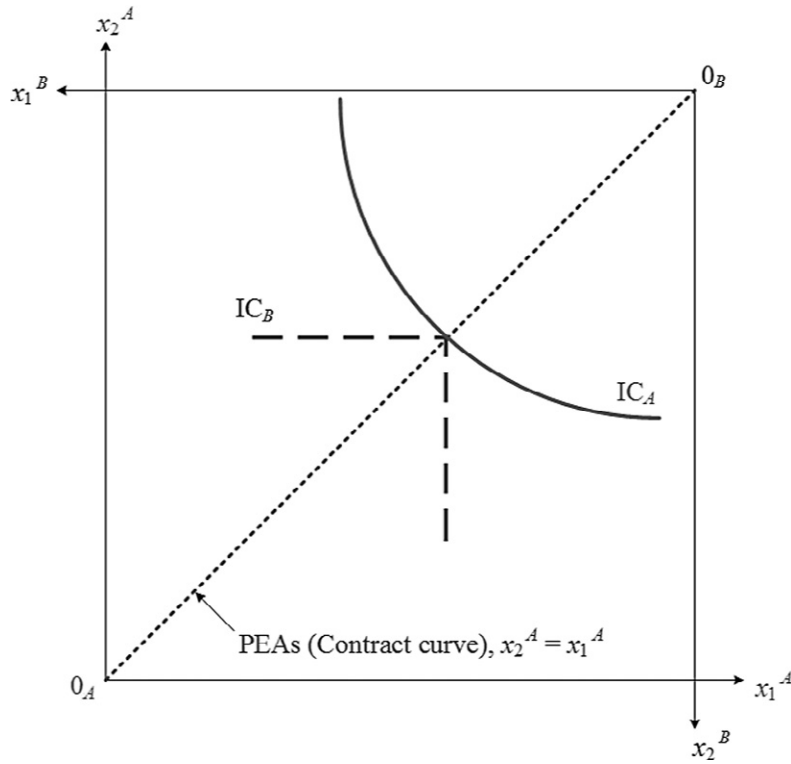


Figure 6.21
Contract curve

$$p_1 x_1^A + x_2^A = 3p_1 + 1,$$

where λ denotes the Lagrange multiplier. Combining the first two equations yields

$$\lambda = \frac{x_2^A}{p_1} = x_1^A, \quad \text{or} \quad p_1 = \frac{x_2^A}{x_1^A}.$$

In addition, for Pareto efficiency, we know that $x_2^A = x_1^A$ must hold, implying that $p_1 = x_2^A / x_1^A = 1$. All that remains is to substitute both the price and the PEA requirement back into the budget constraint, which obtains

$$2x_1^A = 4 \Rightarrow x_1^{A,*} = x_2^{A,*} = 2.$$

Then, using our feasibility conditions, we have

$$\underline{2} + x_1^B = 4 \Rightarrow x_1^{B,*} = x_2^{B,*} = 2,$$

and thus our WEA is $(x_1^{A,*}, x_2^{A,*}; x_1^{B,*}, x_2^{B,*}; p_1/p_2) = (2, 2; 2, 2; 1)$, as shown in figure 6.22.

There are several possible allocations in which $\hat{x}^A$ and $\hat{x}^B$ are the equilibrium allocations, such as $\hat{e}^A = (2, 0)$ and $\hat{e}^B = (2, 4)$. More generally, any allocation satisfying $\hat{e}_1^B + \hat{e}_2^B = 6$ and $\hat{e}_1^A + \hat{e}_2^A = 2$ will give this solution (assuming that the total amount of each good is still 4). ■

6.4.3 Equilibrium with Production

Let us now extend our previous results to a setting where firms are also active. Specifically, assume J firms in the economy, each with production set Y^j that satisfies (1) possible inaction, $0 \in Y^j$ as depicted in the origin of figure 6.23; (2) closed and bounded Y^j , so that points on the production frontier are part of the production set and are thus feasible; and (3) strictly convex Y^j , whereby linear combinations of two production plans also belong to the production set, as depicted in the interior of figure 6.23.

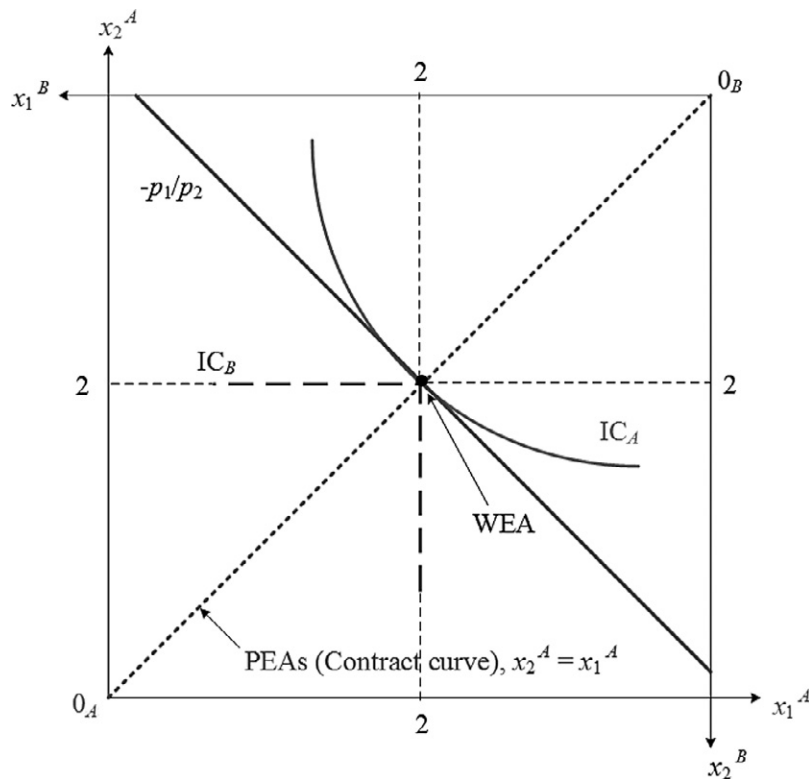


Figure 6.22
Second welfare theorem: WEA and PEAs

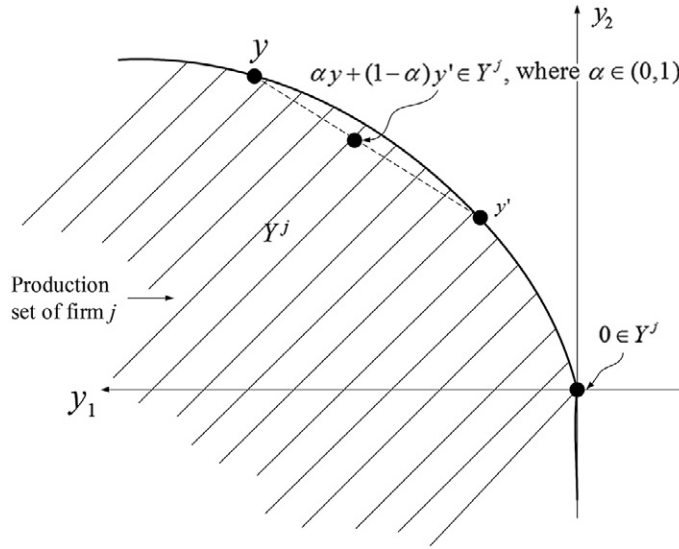


Figure 6.23
Production set Y^j for a representative firm

Similarly to consumers, who independently and simultaneously solve their own UMPs (when facing a fixed price vector $\mathbf{p} \gg \mathbf{0}$), every firm j facing a fixed price vector $\mathbf{p} \gg \mathbf{0}$ independently and simultaneously solves its PMP:

$$\max_{y^j \in Y^j} \mathbf{p} \cdot y^j.$$

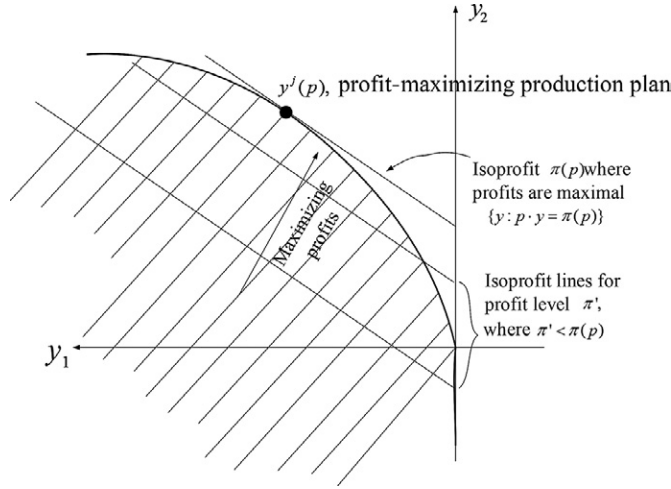
From the above assumptions on production sets Y^j , a profit-maximizing production plan $y^j(\mathbf{p})$ exists for every firm j , and it is unique, as illustrated in figure 6.24 (for more details, see chapter 4 on production theory). In addition, by the theorem of the maximum, both the argmax, $y^j(\mathbf{p})$, and the value function, $\pi^j(\mathbf{p}) \equiv \mathbf{p} \cdot y^j(\mathbf{p})$, are continuous in $\mathbf{p}$.¹⁴

Aggregate Production Set

We can now define the aggregate production set as the sum of all the J firms' production plans (whether profit maximizing or not), which we express mathematically as follows:

$$Y = \left\{ \mathbf{y} \mid \mathbf{y} = \sum_{j=1}^J y^j, \text{ where } y^j \in Y^j \right\}.$$

¹⁴ Since profits are $\pi^j = p_2 y_2 - p_1 y_1$, solving for y_2 yields the isoprofit line $y_2 = \pi^j / p_2 + (p_1 / p_2) y_1$, where π^j / p_2 is the vertical intercept of the isoprofit lines in figure 6.11, while p_1 / p_2 represents their slope.

**Figure 6.24**

Case where $y^j(\mathbf{p})$ exists and is unique

What is the relationship between the production plan $y^j(\mathbf{p})$ that maximizes the individual profits of every firm j and the production plan $\mathbf{y}(\mathbf{p}) \in Y$ that maximizes aggregate profits (i.e., one point in the aggregate production set Y we just defined)? As shown in chapter 4, we can express such a profit-maximizing production plan $\mathbf{y}(\mathbf{p})$ as the sum of each firm's profit-maximizing plans, $\mathbf{y}(\mathbf{p}) = y^1(\mathbf{p}) + y^2(\mathbf{p}) + \dots y^J(\mathbf{p})$.

In this economy with J firms, each firm earns $\pi^j(\mathbf{p})$ profits in equilibrium. How are profits distributed? We can assume that each individual i owns a share θ_{ij} of firm j 's profits, where $0 \leq \theta_{ij} \leq 1$, and that firm j 's profits are distributed across all I consumers, that is, $\sum_{i=1}^I \theta_{ij} = 1$. Note that such distribution of profits allows for multiple sharing profiles, from $\theta_{ij} = 1$ where individual i owns all shares of firm j , to $\theta_{ij} \leq 1/I$, so the shares on firm j are evenly distributed across all I consumers. In this context, consumer i 's budget constraint becomes

$$\mathbf{p} \cdot \mathbf{x}^i \leq \mathbf{p} \cdot \mathbf{e}^i + \sum_{j=1}^J \theta_{ij} \pi^j(\mathbf{p}),$$

where only the last term, $\sum_{j=1}^J \theta_{ij} \pi^j(\mathbf{p})$, is new relative to the standard budget constraint that we considered so far in economies without production. For compactness, we can express the budget constraint as

$$\mathbf{p} \cdot \mathbf{x}^i \leq \mathbf{p} \cdot \mathbf{e}^i + \sum_{j=1}^J \theta_{ij} \pi^j(\mathbf{p}) \Rightarrow \mathbf{p} \cdot \mathbf{x}^i \leq m^i(\mathbf{p}),$$

where $m^i(\mathbf{p})$ denotes all the resources of individual i , either originating from the market value of his initial endowment (if he sells it at the current market prices) or the profits he makes from firms' shares. Last, note that given our previous assumptions on the production sets Y^j , the profit-maximizing plans entail a positive profit for every firm j , implying that $m^i(\mathbf{p}) > 0$.

Equilibrium with Production

In order to characterize equilibrium in economies with production, let us follow a similar approach to that in economies without production, where we started by defining excess demand functions and subsequently used such a definition to compactly identify the set of equilibrium allocations:

Excess demand The excess demand function for good k is

$$z_k(\mathbf{p}) \equiv \sum_{i=1}^I x_k^i(\mathbf{p}, m^i(\mathbf{p})) - \sum_{i=1}^I e_k^i - \underbrace{\sum_{j=1}^J y_k^j(\mathbf{p})}_{\text{New}},$$

where $\sum_{j=1}^J y_k^j(\mathbf{p})$ is a new term relative to our analysis of general equilibrium without production, and denotes the sum of profit-maximizing production of good k across all J firms.

Hence the aggregate excess demand vector is

$$\mathbf{z}(\mathbf{p}) \equiv (z_1(\mathbf{p}), z_2(\mathbf{p}), \dots, z_n(\mathbf{p})).$$

We can now use $\mathbf{z}(\mathbf{p})$ to define a WEA with production.

WEA with production If the price vector is strictly positive in all of its components, $\mathbf{p}^* \gg 0$, a pair of consumption and production bundles $(\mathbf{x}(\mathbf{p}^*), \mathbf{y}(\mathbf{p}^*))$ is a WEA if

1. each consumer i solves his UMP, which becomes the i th entry of $\mathbf{x}(\mathbf{p}^*)$, that is, $\mathbf{x}^i(\mathbf{p}^*, m^i(\mathbf{p}^*))$;
2. each firm j solves its PMP, which becomes the j th entry of $\mathbf{y}(\mathbf{p}^*)$, that is, $\mathbf{y}^j(\mathbf{p}^*)$; and
3. demand equals supply

$$\sum_{i=1}^I \mathbf{x}^i(\mathbf{p}^*, m^i(\mathbf{p}^*)) = \sum_{i=1}^I \mathbf{e}^i + \sum_{j=1}^J \mathbf{y}^j(\mathbf{p}^*),$$

which states the market-clearing condition (or feasibility when expressed for any price vector $\mathbf{p}$).

From every consumer i simultaneously solving his UMP, we obtain that the marginal rate of substitution between any goods 1 and 2 satisfies $MRS_{1,2}^i = MRS_{1,2}^j = p_1/p_2$ for every $j \neq i$ (we also found this result in the barter economies analyzed in previous sections). Similarly, from every firm j simultaneously solving its PMP, we obtain that

$$p_1 F_{1K} = r \quad \text{and} \quad p_1 F_{1L} = w \quad \text{for firm 1, and}$$

$$p_2 F_{2K} = r \quad \text{and} \quad p_2 F_{2L} = w \quad \text{for firm 2.}$$

Dividing these expressions yields

$$MRTS_{L,K}^1 \equiv \frac{F_{1L}}{F_{1K}} = \frac{w}{r} \quad \text{for firm 1,}$$

$$MRTS_{L,K}^2 \equiv \frac{F_{2L}}{F_{2K}} = \frac{w}{r} \quad \text{for firm 2.}$$

Therefore $MRTS_{L,K}^1 = MRTS_{L,K}^2$. Similarly we can divide $p_1 F_{1K} = r$ and $p_2 F_{2K} = r$ to find

$$\frac{p_1 F_{1K}}{p_2 F_{2K}} = \frac{r}{r} = 1,$$

or after rearranging,

$$\frac{p_1}{p_2} = \frac{F_{2K}}{F_{1K}} \equiv MRT_{1,2}^K.$$

A similar result emerges when we divide $p_1 F_{1L} = w$ and $p_2 F_{2L} = w$, that is, $p_1/p_2 = F_{2L}/F_{1L} \equiv MRT_{1,2}^L$. Overall, combining the equilibrium conditions for every consumer i , $MRS_{1,2}^i = p_1/p_2$, and for every input $m = \{K, L\}$, $p_1/p_2 = MRT_{1,2}^m$, yields

$$MRS_{1,2}^i = MRT_{1,2}^m = \frac{p_1}{p_2},$$

where $MRS_{1,2}^i \equiv MU_1^i/MU_2^i$ is decreasing in good 1 (as x_1 increases, MU_1^i decreases while MU_2^i decreases). In contrast, $MRT_{1,2}^m \equiv F_{2m}/F_{1m}$ is decreasing in good 1. Intuitively, in order to increase x_1 , we need to move units of input m from firm 2 to firm 1, thus increasing the marginal product of this input for firm 2 and lowering it for firm 1. Figure 6.25 plots $MRS_{1,2}^i$ and $MRT_{1,2}^m$ as a function of x_1 , crossing each other at the equilibrium level of good 1, x_1^* , at a height of p_1/p_2 .

Existence

Assume that consumers' utility functions are continuous, strictly increasing, and strictly quasi-concave (as considered in previous sections), and that they are initially

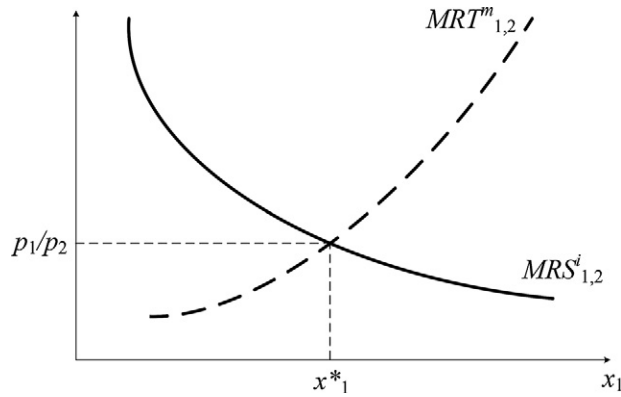


Figure 6.25
Equilibrium with production

endowed with positive units of at least one good, so the sum $\sum_{i=1}^I \mathbf{e}^i \gg 0$. Additionally every firm j 's production set Y^j is closed and bounded,¹⁵ strictly convex, and satisfies the property of inaction being possible (as stated at the beginning of this section). In this economy, there is a price vector $\mathbf{p}^* \gg 0$ such that a WEA exists, that is $z(\mathbf{p}^*) = 0$. (For a proof of this result, see Varian 1992.) Let us now consider a numerical example of equilibrium with production with two consumers and two firms.

Example 6.8: Finding WEAs with production Consider a two-consumer, two-good economy where every consumer $i = \{1, 2\}$ has utility function $u^i = x_1^i x_2^i$. There are two firms in this economy, each using capital and labor as inputs to produce one of the consumption goods. Firm 1 produces good 1 with production function $y_1 = K_1^{0.75} L_1^{0.25}$, and firm 2 produces good 2 with production function $y_2 = K_2^{0.25} L_2^{0.75}$. Consumer A is endowed with $(K^A, L^A) = (1, 1)$, while consumer 2 is endowed with $(K^B, L^B) = (2, 1)$. Let us find a WEA in this economy with production.

UMPs

Starting on the consumer side of this problem, consumer A 's utility maximization problem is

$$\begin{aligned} \max_{x_1^A, x_2^A} \quad & x_1^A x_2^A \\ \text{subject to} \quad & p_1 x_1^A + p_2 x_2^A = rK^A + wL^A, \end{aligned}$$

¹⁵ Recall from chapter 4 that bounded production sets allow for efficient production plans (graphically, those on the production frontier).

where r and w are the prices for capital and labor, respectively. Taking first-order conditions gives us the tangency condition for utility maximization under regular preferences

$$\frac{p_1}{p_2} = MRS_{1,2}^A \Rightarrow \frac{p_1}{p_2} = \frac{x_2^A}{x_1^A} \Rightarrow p_1 x_1^A = p_2 x_2^A \quad (6.1)$$

Likewise consumer B 's utility maximization problem is

$$\max_{x_1^B, x_2^B} x_1^B x_2^B$$

$$\text{subject to } p_1 x_1^B + p_2 x_2^B = rK^B + wL^B,$$

with first-order conditions giving

$$\frac{p_1}{p_2} = MRS_{1,2}^B \Rightarrow \frac{p_1}{p_2} = \frac{x_2^B}{x_1^B} \Rightarrow p_1 x_1^B = p_2 x_2^B. \quad (6.2)$$

Now, taking equations (6.1) and (6.2) and adding them together, we have

$$p_1(x_1^A + x_1^B) = p_2(x_2^A + x_2^B).$$

But recall that $x_1^A + x_1^B$ is the left side of our feasibility condition, so $x_1^A + x_1^B = y_1 = K_1^{0.75} L_1^{0.25}$. Substituting both feasibility conditions into the above equation and rearranging gives

$$\frac{p_1}{p_2} = \frac{K_2^{0.25} L_2^{0.75}}{K_1^{0.75} L_1^{0.25}}. \quad (6.3)$$

PMPs

Next we move to the production side of the economy. Firm 1's PMP is

$$\max_{K_1, L_1} p_1 K_1^{0.75} L_1^{0.25} - rK_1 - wL_1$$

with first-order conditions

$$r = 0.75 p_1 K_1^{-0.25} L_1^{0.25},$$

$$w = 0.25 p_1 K_1^{0.75} L_1^{-0.75}.$$

Combining these expressions gives the tangency condition for profit maximization under regular technologies:

$$\frac{r}{w} = MRTS_{L,K}^1 \Rightarrow \frac{r}{w} = 3 \frac{L_1}{K_1}. \quad (6.4)$$

Likewise firm 2's PMP gives the following first-order conditions

$$r = 0.25 p_2 K_2^{-0.75} L_2^{0.75},$$

$$w = 0.75 p_2 K_2^{0.25} L_2^{-0.25},$$

and combining them gives the tangency condition

$$\frac{r}{w} = MRTS_{L,K}^2 \Rightarrow \frac{r}{w} = \frac{1}{3} \frac{L_2}{K_2}. \quad (6.5)$$

Combining both *MRTS* gives

$$3 \frac{L_1}{K_1} = \frac{1}{3} \frac{L_2}{K_2} \Rightarrow \frac{K_1}{L_1} = 9 \frac{K_2}{L_2}.$$

Intuitively, this result implies that firm 1 is more capital intensive than firm 2, since its capital-to-labor ratio is higher. Solving for the price of capital, r , in the above expressions and setting them equal to each other gives

$$0.75 p_1 K_1^{-0.25} L_1^{0.25} = 0.25 p_2 K_2^{-0.75} L_2^{0.75} \Rightarrow \frac{p_1}{p_2} = \frac{1}{3} \left(\frac{K_1}{L_1} \right)^{0.25} \left(\frac{K_2}{L_2} \right)^{-0.75}, \quad (6.6)$$

and likewise, setting both firms' price of labor equal to each other gives

$$0.25 p_1 K_1^{0.75} L_1^{-0.75} = 0.75 p_2 K_2^{0.25} L_2^{-0.25} \Rightarrow \frac{p_1}{p_2} = 3 \left(\frac{K_1}{L_1} \right)^{-0.75} \left(\frac{K_2}{L_2} \right)^{0.25}. \quad (6.7)$$

Now setting (6.3)—from both consumers' UMPs—equal to (6.6)—from both firms' PMPs—gives

$$\frac{K_2^{0.25} L_2^{0.75}}{K_1^{0.75} L_1^{0.25}} = \frac{1}{3} \left(\frac{K_1}{L_1} \right)^{0.25} \left(\frac{K_2}{L_2} \right)^{-0.75} \Rightarrow K_1 = 3 K_2,$$

and by our feasibility conditions, we know that $K_1 + K_2 = K^A + K^B = 3$, or $K_2 = 3 - K_1$. Substituting, we find the profit-maximizing demands for capital use by firms 1 and 2,

$$K_1 = 3(3 - K_1) \Rightarrow K_1^* = \frac{9}{4},$$

$$K_2^* = \frac{1}{3} K_1^* = \frac{3}{4}.$$

Next we set (6.3)—from both consumers' UMPs—equal to (6.7)—from both firms' PMPs—as

$$\frac{K_2^{0.25} L_2^{0.75}}{K_1^{0.75} L_1^{0.25}} = 3 \left(\frac{K_1}{L_1} \right)^{-0.75} \left(\frac{K_2}{L_2} \right)^{0.25} \Rightarrow L_1 = \frac{1}{3} L_2,$$

and by our feasibility condition, we know that $L_1 + L_2 = L^A + L^B = 2$, or $L_2 = 2 - L_1$. Substituting, we find the labor demands for firm 1 and 2,

$$L_1 = \frac{1}{3}(2 - L_1) \Rightarrow L_1^* = \frac{1}{2},$$

$$L_2^* = 3L_1^* = \frac{3}{2}.$$

From here, we can substitute these values into equation (6.3) to find that the equilibrium price ratio is

$$\frac{p_1}{p_2} = \frac{(3/4)^{0.25} (3/2)^{0.75}}{(9/4)^{0.75} (1/2)^{0.25}} = \sqrt{2/3},$$

and normalizing the price of good 2, $p_2 = 1$, gives $p_1 = \sqrt{2/3} \cong 0.82$. Furthermore we can substitute our calculated values into the price of capital and labor to find

$$r^* = 0.75(0.82) \left(\frac{9}{4} \right)^{-0.25} \left(\frac{1}{2} \right)^{0.25} = 0.42,$$

$$w^* = 0.25(0.82) \left(\frac{9}{4} \right)^{0.75} \left(\frac{1}{2} \right)^{-0.75} = 0.63.$$

Last, we return to the consumer side of the market. Using consumer A 's tangency condition (equation 6.1), we know that

$$x_2^A = \frac{p_1}{p_2} x_1^A \Rightarrow x_2^A = 0.82 x_1^A,$$

and substituting this value into consumer A 's budget constraint gives

$$p_1 x_1^A + p_2 (0.82 x_1^A) = r K^A + w L^A.$$

Inserting our calculated values for p_1 , p_2 , r , k^A , w and l^A , and solving this expression for x_1^A yields

$$x_1^{A,*} = 0.64,$$

$$x_2^{A,*} = 0.82 x_1^{A,*} = 0.53.$$

Performing the same process with the tangency condition of consumer B (equation 6.2) yields

$$x_1^{B,*} = 0.90$$

$$x_2^{B,*} = 0.74 ,$$

which completes our WEA in this economy with production:

$$(x_1^A, x_2^A; x_1^B, x_2^B; \frac{P_1}{P_2}; L_1, L_2, K_1, K_2) = \left(0.64, 0.53; 0.90, 0.74; 0.82; \frac{1}{2}, \frac{3}{2}, \frac{9}{4}, \frac{3}{4} \right). \blacksquare$$

Equilibrium with Production—Welfare

In this subsection we seek to extend the first and second welfare theorems to economies with production, connecting WEA and PEAs. Before stating the first welfare theorem, let us define what we mean by a PEA in economies with production.

Pareto efficiency A feasible allocation $(\mathbf{x}, \mathbf{y})$ is Pareto efficient if there is no other feasible allocation $(\bar{\mathbf{x}}, \bar{\mathbf{y}})$ such that

$$u^i(\bar{\mathbf{x}}^i) \geq u^i(\mathbf{x}^i)$$

for every consumer $i \in I$, with $u^i(\bar{\mathbf{x}}^i) > u^i(\mathbf{x}^i)$ for at least one consumer i .

That is, a feasible allocation of bundles to consumers and production plans to firms is Pareto efficient if there is no other feasible allocation that makes at least one consumer strictly better off and no consumer worse off.

As in section 6.4.1 analyzing barter economies, let us describe how to mathematically find the set of PEAs. In particular, in an economy with two goods, two consumers, two firms, and two inputs (labor and capital), the set of PEAs solves

$$\max_{x_1^1, x_2^1, x_1^2, x_2^2, x_1^1, x_2^1, x_1^2, x_2^2, L_1, K_1, L_2, K_2 \geq 0} u^1(x_1^1, x_2^1)$$

$$\text{subject to } u^2(x_1^2, x_2^2) \geq \bar{u}^2,$$

$x_1^1 + x_1^2 \leq F_1(L_1, K_1)$ and $x_2^1 + x_2^2 \leq F_2(L_2, K_2)$ (technological feasibility), and $L_1 + L_2 \leq \bar{L}$ and $K_1 + K_2 \leq \bar{K}$ (input feasibility).

The Lagrangian associated with this maximization problem is

$$\begin{aligned}\mathcal{L} = & u^1(x_1^1, x_2^1) + \lambda[u^2(x_1^2, x_2^2) - \bar{u}^2] + \mu_1[F_1(L_1, K_1) - x_1^1 - x_1^2] \\ & + \mu_2[F_2(L_2, K) - x_2^1 - x_2^2] + \delta_L[\bar{L} - L_1 - L_2] + \delta_K[\bar{K} - K_1 - K_2].\end{aligned}$$

In the case of interior solutions, the set of first-order conditions yields a condition for efficiency in consumption that we also found in barter economies, $MRS_{1,2}^1 = MRS_{1,2}^2$. The first-order conditions with respect to inputs L_j and K_j yield a condition for efficiency that we encountered in the chapter on production theory,

$$\frac{\partial F_j / \partial L}{\partial F_j / \partial K} = \frac{\partial F_m / \partial L}{\partial F_m / \partial K} \quad \text{for every two firms } j \neq m.$$

That is, the marginal rate of technical substitution between inputs, $MRTS_{L,K}$, must coincide across firms. Otherwise, welfare could be increased by assigning more labor to the firm with the highest $MRTS_{L,K}$. Finally, combining the two conditions above for efficiency in consumption and production, we obtain

$$\frac{\partial U^i / \partial x_1^i}{\partial U^i / \partial x_2^i} = \frac{\partial F_2 / \partial L}{\partial F_1 / \partial L}.$$

Or, more compactly, $MRS_{1,2}^i$ must coincide with the rate at which firms can transform units of good 1 into units of good 2, namely the marginal rate of transformation $MRT_{1,2}$.¹⁶ Intuitively, for an allocation to be efficient, we need that the rate at which consumers are willing to substitute goods 1 and 2 coincides with the rate at which good 1 can be transformed into good 2. We can now use this definition of Pareto efficiency to state the first welfare theorem in economies with production.

First welfare theorem with production If the utility function of every individual i , u^i , is strictly increasing, then every WEA is Pareto efficient.

Proof We prove this result by contradiction. In particular, suppose that $(\mathbf{x}, \mathbf{y})$ is a WEA at prices $\mathbf{p}^*$ but is *not* Pareto efficient. Because $(\mathbf{x}, \mathbf{y})$ is a WEA, it is feasible, that is,

$$\sum_{i=1}^I \mathbf{x}^i = \sum_{i=1}^I \mathbf{e}^i + \sum_{j=1}^J \mathbf{y}^j. \quad (A)$$

¹⁶ Indeed, if we move labor from firm 2 to firm 1, the production of good 2 increases by $\partial F_2 / \partial L$ while that of good 1 decreases by $\partial F_1 / \partial L$. Hence, in order to increase the total output of good 1 by one unit, we need $(\partial F_2 / \partial L) / (\partial F_1 / \partial L)$ units of good 2.

In addition, because $(\mathbf{x}, \mathbf{y})$ is *not* Pareto efficient, there exists some other feasible allocation $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ such that $u^i(\hat{\mathbf{x}}^i) \geq u^i(\mathbf{x}^i)$ for every consumer $i \in I$, with $u^i(\hat{\mathbf{x}}^i) > u^i(\mathbf{x}^i)$ for at least one consumer. That is, the alternative allocation $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ makes at least one consumer strictly better off than WEA $(\mathbf{x}, \mathbf{y})$ without making others worse off. Since utility function $u^i(\cdot)$ is increasing, this implies that bundle $\hat{\mathbf{x}}^i$ contains more of at least one good than $\mathbf{x}^i$, and thus is more costly than $\mathbf{x}^i$, meaning $\mathbf{p}^* \cdot \hat{\mathbf{x}}^i \geq \mathbf{p}^* \cdot \mathbf{x}^i$ for every individual i (with at least one strict inequality). Summing over all consumers yields

$$\mathbf{p}^* \cdot \sum_{i=1}^I \hat{\mathbf{x}}^i > \mathbf{p}^* \cdot \sum_{i=1}^I \mathbf{x}^i. \quad (B)$$

Combining inequalities *A* and *B* with the feasibility of allocation yields

$$\mathbf{p}^* \cdot \left(\sum_{i=1}^I \mathbf{e}^i + \sum_{j=1}^J \hat{\mathbf{y}}^j \right) > \mathbf{p}^* \cdot \left(\sum_{i=1}^I \mathbf{e}^i + \sum_{j=1}^J \mathbf{y}^j \right),$$

or, after rearranging,

$$\mathbf{p}^* \cdot \sum_{j=1}^J \hat{\mathbf{y}}^j > \mathbf{p}^* \cdot \sum_{j=1}^J \mathbf{y}^j.$$

However, this result implies that $\mathbf{p}^* \cdot \hat{\mathbf{y}}^j > \mathbf{p}^* \cdot \mathbf{y}^j$ for some firm j , indicating that production plan $\mathbf{y}^j$ was *not* profit maximizing and, as a consequence, it cannot be part of a WEA. We therefore reached a contradiction, implying that the original statement was true: if an allocation $(\mathbf{x}, \mathbf{y})$ is a WEA, it must also be Pareto efficient. ■

Example 6.9: WEA and PEA with production Consider the setting described in example 6.8. The set of PEAs must satisfy

$$MRS_{1,2}^A = MRS_{1,2}^B \quad \text{and} \quad MRTS_{L,K}^1 = MRTS_{L,K}^2.$$

From equation (6.1) we have

$$MRS_{1,2}^A = \frac{x_2^A}{x_1^A} = \frac{0.53}{0.64} = 0.82,$$

and from equation (6.2),

$$MRS_{1,2}^B = \frac{x_2^B}{x_1^B} = \frac{0.74}{0.90} = 0.82,$$

which implies that $MRS_{1,2}^A = MRS_{1,2}^B$. Likewise from equation (6.4) we have

$$MRTS_{L,K}^1 = 3 \frac{L_1}{K_1} = 3 \frac{1/2}{9/4} = \frac{2}{3},$$

and from equation (6.5),

$$MRTS_{L,K}^2 = \frac{1}{3} \frac{L_2}{K_2} = \frac{1}{3} \frac{3/2}{3/4} = \frac{2}{3},$$

which implies that $MRTS_{L,K}^1 = MRTS_{L,K}^2$. Since both of these conditions hold, our WEA from example 6.8 is Pareto efficient. ■

Second welfare theorem with production Consider, again, the assumptions on consumers and producers described above. For every PEA $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ we can now find: (1) a profile of income transfers $(T_1, T_2, \dots, T_I)$ redistributing income among consumers, satisfying $\sum_{i=1}^I T_i = 0$, and (2) a price vector $\bar{\mathbf{p}}$, such that:

1. Bundle $\hat{\mathbf{x}}^i$ solves the UMP

$$\max_{\mathbf{x}^i} u^i(\mathbf{x}^i) \quad \text{subject to} \quad \bar{\mathbf{p}} \cdot \mathbf{x}^i \leq m^i(\bar{\mathbf{p}}) + T_i \quad \text{for every individual } i \in I,$$

where individual i 's original income $m^i(\bar{\mathbf{p}})$ is increased (if the transfer he receives T_i is positive) or decreased (if it is negative).

2. Production plan $\hat{\mathbf{y}}^j$ solves the PMP

$$\max_{\mathbf{y}^j} \bar{\mathbf{p}} \cdot \mathbf{y}^j \quad \text{subject to} \quad \mathbf{y}^j \in Y^j \quad \text{for every firm } j \in J.$$

Example 6.10: Second welfare theorem with production Consider an alternative allocation in the set of PEAs identified in example 6.9, such as $(\hat{x}_1^A, \hat{x}_2^A; \hat{x}_1^B, \hat{x}_2^B) = (0.75, 0.61; 0.79, 0.65)$. This allocation can be a WEA with the appropriate transfers. Consumer A 's budget constraint becomes

$$p_1 \hat{x}_1^A + p_2 \hat{x}_2^A = rK^A + wL^A + T_1.$$

Recall that $(p_1, p_2; K^A, L^A; r, w) = (0.82, 1; 1, 1; 0.42, 0.63)$ remains unchanged from our WEA in example 6.8 due to our allocation being Pareto efficient. Substituting these values into consumer A 's budget constraint gives

$$0.82 \hat{x}_1^A + \hat{x}_2^A = 1.05 + T_1;$$

also recall that

$$\frac{p_1}{p_2} = \frac{\hat{x}_2^A}{\hat{x}_1^A} \Rightarrow \hat{x}_2^A = 0.82\hat{x}_1^A.$$

After substituting these results into the equation above, we obtain

$$2(0.82)\underbrace{(0.75)}_{\hat{x}_1^A} = 1.05 + T_1 \Rightarrow T_1 = 0.17.$$

Likewise for consumer B his budget constraint becomes

$$p_1\hat{x}_1^B + p_2\hat{x}_2^B = rK^B + wL^B + T_2,$$

and after substituting our unchanged values $(p_1, p_2; K^B, L^B; r, w) = (0.82, 1; 2, 1; 0.42, 0.63)$, we obtain

$$0.82\hat{x}_1^B + \hat{x}_2^B = 1.47 + T_2.$$

Similarly, for consumer B , we can write

$$\frac{p_1}{p_2} = \frac{\hat{x}_2^B}{\hat{x}_1^B} \Rightarrow \hat{x}_2^B = 0.82\hat{x}_1^B,$$

and then, by substituting, obtain

$$2(0.82)\underbrace{(0.79)}_{\hat{x}_1^B} = 1.47 + T_2 \Rightarrow T_2 = -0.17.$$

Clearly, $T_1 + T_2 = 0$, and thus these transfers allow for our new allocation to be a WEA. ■

6.5 Comparative Statics

In this section we briefly explore some comparative statics of our equilibrium results at the WEA in economies with production. Specifically, we analyze how equilibrium outcomes are affected by an increase in the price of one good, and then by an increase in the endowment of one input. For both questions, consider a setting with two goods, each being produced by two factors, 1 and 2, under constant returns to scale (CRS). Given CRS, a necessary condition for input prices (w_1^*, w_2^*) to be in equilibrium is that firms produce until their marginal costs equal the price of the good, that is,

$$c_1(w_1, w_2) = p_1 \quad \text{and} \quad c_2(w_1, w_2) = p_2.$$

For compactness, let $z_{1j}(w)$ denote firm j 's demand for factor 1, and $z_{2j}(w)$ be its demand for factor 2.¹⁷ Hence we say that the production of good 1 is *relatively more intensive* in factor 1 than is the production of good 2 if

$$\frac{z_{11}(w)}{z_{21}(w)} > \frac{z_{12}(w)}{z_{22}(w)},$$

where $z_{1j}(w)/z_{2j}(w)$ represents firm j 's demand for input 1 relative to that of input 2.

6.5.1 Changes in the Price of One Good p_j (Stolper–Samuelson Theorem)¹⁸

Consider an economy with two consumers and two firms satisfying the factor intensity assumption given above. If the price of good j , p_j , increases, then (1) the equilibrium price of the factor more intensively used in the production of good j increases, while (2) the equilibrium price of the other factor decreases.

Proof Let us first take the equilibrium conditions about marginal costs being equal to output prices:

$$c_1(w_1, w_2) = p_1 \quad \text{and} \quad c_2(w_1, w_2) = p_2.$$

Differentiating with respect to both input prices, we obtain

$$\begin{aligned} \frac{\partial c_1(w_1, w_2)}{\partial w_1} dw_1 + \frac{\partial c_1(w_1, w_2)}{\partial w_2} dw_2 &= dp_1, \\ \frac{\partial c_2(w_1, w_2)}{\partial w_1} dw_1 + \frac{\partial c_2(w_1, w_2)}{\partial w_2} dw_2 &= dp_2. \end{aligned}$$

Applying Shephard's lemma, $\partial c_i(w_1, w_2)/\partial w_j = z_{ij}(w)$ results in

$$z_{11}(w)dw_1 + z_{12}(w)dw_2 = dp_1,$$

$$z_{21}(w)dw_1 + z_{22}(w)dw_2 = dp_2.$$

Hence, if only price p_1 varies, then $dp_2 = 0$. We can rewrite the second expression as $dw_2 = (-z_{21}/z_{22})/dw_1$. We can now use the first expression. In particular, solving for dw_1/dp_1 yields

¹⁷ This is equivalent to the factor demand correspondences $z(w, q)$ in the chapter on production theory where, for simplicity, we consider the production of one unit of output $q=1$, which helps us ignore the second argument of $z(w, q)$.

¹⁸ See Stolper and Samuelson (1941).

$$\frac{dw_1}{dp_1} = \frac{z_{22}}{z_{11}z_{22} - z_{12}z_{21}}.$$

Solving, instead, for dw_2/dp_1 yields

$$\frac{dw_2}{dp_1} = -\frac{z_{21}}{z_{11}z_{22} - z_{12}z_{21}}$$

From the factor intensity condition, $z_{11}(w)/z_{21}(w) > z_{12}(w)/z_{22}(w)$, we know that $z_{11}z_{22} - z_{12}z_{21} > 0$, which implies the denominator in both dw_1/dp_1 and dw_2/dp_1 must be positive. Hence, since the numerator is also positive (they are just factor demands), the overall sign of the previous expressions is

$$\frac{dw_1}{dp_1} > 0 \quad \text{and} \quad \frac{dw_2}{dp_1} < 0.$$

Intuitively, if the price of good 1 increases, the price of input 1 (the input more intensively used in the production of good 1), w_1 , increases while that of the other input (less intensively used than input 1), w_2 , decreases. ■

Example 6.11: Stolper–Samuelson theorem Returning to our problem in example 6.8, let us now solve for the input demands:

$$r_1 = p_1 0.75 K_1^{-0.25} L_1^{0.25} \Rightarrow z_{11} \equiv K_1 = \left(\frac{0.75 p_1}{r} \right)^4 L_1,$$

$$w_1 = p_1 0.25 K_1^{0.75} L_1^{-0.75} \Rightarrow z_{21} \equiv L_1 = \left(\frac{p_1}{4w} \right)^{4/3} K_1,$$

$$r_2 = p_2 0.25 K_2^{-0.75} L_2^{0.75} \Rightarrow z_{12} \equiv K_2 = \left(\frac{p_2}{4r} \right)^{4/3} L_2,$$

$$w_2 = p_2 0.75 K_2^{0.25} L_2^{-0.25} \Rightarrow z_{22} \equiv L_2 = \left(\frac{0.75 p_2}{w} \right)^4 K_2.$$

Since firm 1 is more capital intensive than firm 2, $z_{11}z_{22} - z_{12}z_{21} > 0$ must hold, that is,

$$\left(\frac{0.75 p_1}{r} \right)^4 L_1 \left(\frac{0.75 p_2}{w} \right)^4 K_2 - \left(\frac{p_1}{4w} \right)^{4/3} K_1 \left(\frac{p_2}{4r} \right)^{4/3} L_2 > 0.$$

Recall from example 6.8 that $K_1/L_1 = 9(K_2/L_2)$ or, rearranging, that $K_1/L_2 = 9K_2/L_1$. Substituting these values and simplifying the expression above gives

$$36.33 \left(\frac{p_1 p_2}{r w} \right)^{8/3} - 1 > 0 \Rightarrow \frac{p_1 p_2}{r w} > 1.16$$

and in our solution, $p_1 p_2 / r w = 3.08$; hence condition $z_{11} z_{22} - z_{12} z_{21} > 0$ holds.

Next, observe that both z_{22} and z_{11} are trivially positive. We can apply the Stolper–Samuelson theorem at this point to find

$$\frac{dw_1}{dp_1} = \frac{z_{22}}{z_{11} z_{22} - z_{12} z_{21}} > 0,$$

$$\frac{dw_2}{dp_1} = - \frac{z_{21}}{z_{11} z_{22} - z_{12} z_{21}} < 0. \blacksquare$$

6.5.2 Changes in Endowments (Rybczynski's Theorem)¹⁹

Let us now examine how the equilibrium output is affected by a change in the endowment of one input. Consider an economy with two consumers and two firms satisfying the factor intensity assumption given above. Additionally assume that this is a small open economy, so output prices are given (no market power). In this setting, if the endowment of a factor increases, the production of the good that uses this factor more intensively increases, whereas the production of the other good decreases.

Proof Consider an economy with two factors, labor and capital, and two goods, 1 and 2. In addition recall that for a firm j , $z_{Lj}(w)$ denotes its factor demand for labor (when producing one unit of output), and similarly $z_{Kj}(w)$ represents its factor demand for capital. Then factor feasibility requires that

$$L = z_{L1}(w) \cdot y_1 + z_{L2}(w) \cdot y_2,$$

where the first (second) term measures the units of labor that firm 1 (2, respectively) demands. A similar condition applies to capital:

$$K = z_{K1}(w) \cdot y_1 + z_{K2}(w) \cdot y_2.$$

Differentiating the first condition with respect to L , we have

$$dL = z_{L1} \cdot \frac{\partial y_1}{\partial L} + z_{L2} \cdot \frac{\partial y_2}{\partial L}.$$

¹⁹ See Rybczynski (1955).

Dividing both sides by the aggregate amount of labor yields

$$\frac{dL}{L} = \frac{z_{L1}}{L} \cdot \frac{\partial y_1}{\partial L} + \frac{z_{L2}}{L} \cdot \frac{\partial y_2}{\partial L}.$$

We multiply the first term on the right-hand side by y_1/y_1 and the second term by y_2/y_2 to obtain

$$\frac{dL}{L} = \frac{z_{L1} \cdot y_1}{L} \cdot \frac{\partial y_1 / \partial L}{y_1} + \frac{z_{L2} \cdot y_2}{L} \cdot \frac{\partial y_2 / \partial L}{y_2},$$

which can be more compactly expressed using $z_{Li}(w) \cdot y_i / L \equiv \gamma_{Li}$, which is the share of labor used by firm i ; $(\partial y_i / \partial L) / y_i \equiv \% \Delta y_i$, which is the percentage increase in the production of firm i brought by the increase in the endowment of labor; and $dL / L \equiv \% \Delta L$, which is the percentage increase in the endowment of labor in the economy. As a consequence the expression above becomes

$$\% \Delta L = \gamma_{L1} \cdot (\% \Delta y_1) + \gamma_{L2} \cdot (\% \Delta y_2).$$

A similar argument with the endowment of capital yields

$$\% \Delta K = \gamma_{K1} \cdot (\% \Delta y_1) + \gamma_{K2} \cdot (\% \Delta y_2).$$

In addition, note that labor shares $\gamma_{L1}, \gamma_{L2} \in (0, 1)$ and that $\gamma_{L1}, \gamma_{L2} = 1$, implying that $\% \Delta L$ is a linear combination of $\% \Delta y_1$ and $\% \Delta y_2$, and a similar argument applies for capital shares $\gamma_{K1}, \gamma_{K2} \in (0, 1)$ and $\% \Delta K$. Finally, since capital is assumed to be more intensively used by firm 1, in that $K_1 / L_1 > K_2 / L_2$, or $\gamma_{K1} > \gamma_{L1}$ for firm 1 and $\gamma_{K2} > \gamma_{L2}$ for firm 2. As a consequence, observing the expression describing the percentage change in the endowment of capital and labor, if capital becomes relatively more abundant than labor, in that $\% \Delta K > \% \Delta L$, then it must be that $\% \Delta y_1$ is larger than $\% \Delta y_2$. That is,

$$\begin{array}{ccccccc} \% \Delta L & = & \gamma_{L1} & \cdot & \underbrace{(\% \Delta y_1)} & + & \gamma_{L2} \cdot \underbrace{(\% \Delta y_2)} \\ \wedge & & \wedge & & \parallel & & \vee & & \parallel \\ \% \Delta K & = & \gamma_{K1} & \cdot & \underbrace{(\% \Delta y_1)} & + & \gamma_{K2} \cdot \underbrace{(\% \Delta y_2)} \end{array}$$

In other words, the change in the input endowment produces a more than proportional increase in the good whose production was more intensive in the use of that input, for example, a 1 percent increase in the capital endowment increases y_1 by more than 1 percent. The converse argument applies for labor and the production of good 2. More compactly, $\% \Delta y_1 > \% \Delta K$ and $\% \Delta L < \% \Delta y_2$.

Alternative proof We can alternatively prove the above result by explicitly finding $\% \Delta y_1$ and $\% \Delta y_2$. Consider again the expressions above describing the increase in the labor and capital endowments:

$$\% \Delta L = \gamma_{L1} \cdot (\% \Delta y_1) + \gamma_{L2} \cdot (\% \Delta y_2),$$

$$\% \Delta K = \gamma_{K1} \cdot (\% \Delta y_1) + \gamma_{K2} \cdot (\% \Delta y_2),$$

or, more compactly,

$$\begin{bmatrix} \% \Delta L \\ \% \Delta K \end{bmatrix} = \begin{bmatrix} \gamma_{L1} & \gamma_{L2} \\ \gamma_{K1} & \gamma_{K2} \end{bmatrix} \cdot \begin{bmatrix} \% \Delta y_1 \\ \% \Delta y_2 \end{bmatrix}.$$

Applying Cramer's rule to obtain $\% \Delta y_1$ and $\% \Delta y_2$, we find that

$$\begin{bmatrix} \% \Delta y_1 \\ \% \Delta y_2 \end{bmatrix} = \begin{bmatrix} \gamma_{L1} & \gamma_{L2} \\ \gamma_{K1} & \gamma_{K2} \end{bmatrix}^{-1} \cdot \begin{bmatrix} \% \Delta L \\ \% \Delta K \end{bmatrix} = \frac{1}{\gamma_{L1}\gamma_{K2} - \gamma_{K1}\gamma_{L2}} \begin{bmatrix} \gamma_{K2} & -\gamma_{L2} \\ -\gamma_{K1} & \gamma_{L1} \end{bmatrix} \begin{bmatrix} \% \Delta L \\ \% \Delta K \end{bmatrix}.$$

In this context, we can define the term in the denominator as $A = \gamma_{L1}\gamma_{K2} - \gamma_{K1}\gamma_{L2}$, which is negative, since firm 1 is more capital intensive than firm 2, given that $\gamma_{K1} > \gamma_{L1}$ and $\gamma_{K2} > \gamma_{L2}$. As a consequence the expression above can be more compactly represented as

$$\% \Delta y_1 = \frac{1}{A} [\gamma_{K2} (\% \Delta L) - \gamma_{L2} (\% \Delta K)],$$

$$\% \Delta y_2 = \frac{1}{A} [-\gamma_{K1} (\% \Delta L) + \gamma_{L1} (\% \Delta K)].$$

For simplicity, consider that only the capital endowment changes, $\% \Delta K > 0$ and $\% \Delta L = 0$, which further simplifies the expressions above to

$$\% \Delta y_1 = -\frac{1}{A} [\gamma_{L2} (\% \Delta K)] > 0,$$

$$\% \Delta y_2 = \frac{1}{A} [\gamma_{L1} (\% \Delta K)] < 0,$$

since $A < 0$, as shown above. Hence $\% \Delta y_1 > 0 > \% \Delta y_2$. In words, as capital endowment increases the production of the capital-intensive good increases, whereas the production of the labor-intensive good decreases. Let us now examine the relative percentage change. Expanding term A in the expression above of $\% \Delta y_1$ yields

$$\% \Delta y_1 = -\frac{\gamma_{L2}}{\gamma_{L1}\gamma_{K2} - \gamma_{K1}\gamma_{L2}} (\% \Delta K) = \frac{\gamma_{L2}}{\gamma_{K1}\gamma_{L2} - \gamma_{L1}\gamma_{K2}} (\% \Delta K).$$

Rearranging, and dividing the right-hand side by γ_{L2} , we obtain

$$\frac{\% \Delta y_1}{\% \Delta K} = \frac{1}{\gamma_{K1} - \gamma_{L1} (\gamma_{K2} / \gamma_{L2})}.$$

Finally, the right-hand side yields a number larger than 1. Indeed, the infimum of the right-hand term occurs when the denominator reaches its highest value, which is at $\gamma_{K1} = 1$ and $\gamma_{L1}(\gamma_{K2} / \gamma_{L2}) = 0$. Hence $\% \Delta y_1 / \% \Delta K > 1$, implying that $\% \Delta y_1 > \% \Delta K$, as required. (A similar proof applies if we expand term A in the expression of $\% \Delta y_2$, yielding $\% \Delta y_2 > \% \Delta L$, which is left to the reader for practice.) ■

Example 6.12: Rybczynski theorem Consider the production decisions of the two firms in example 6.8, where we found that $K_1 = 3K_2$ and $K_1 + K_2 = \bar{K} = 3$. Assume that total endowment of capital increases to $\bar{K} = 5$, which is to say, $K_2 = 5 - K_1$, yielding profit-maximizing demands for capital of

$$K_1 = 3(5 - K_1) \Rightarrow K_1^* = \frac{15}{4},$$

$$K_2 = \frac{1}{3} K_1^* = \frac{5}{4}.$$

Similarly for labor we found that $L_1 = \frac{1}{3} L_2$ and $L_1 + L_2 = \bar{L} = 2$. However, we do not alter the aggregate endowment of labor, $\bar{L} = 2$, as we seek to increase the endowment of the input more intensively used by firm 1 (capital). We have thus shown that capital use by firm 1 increases from $K_1^* = 9/4$ to $15/4$. That is, if firm 1 uses capital more intensively than firm 2 does, meaning $K_1/L_1 > K_2/L_2$, since $(9/4)/(1/2) > (3/4)/(3/2)$, an increase in the endowment of capital of $(5 - 3)/3 = 0.66 = 0.66$ percent entails an increase in good 1's output by 100 percent while that of good 2 decreases by only 33.33 percent.

6.6 Introducing Taxes

6.6.1 Tax on Goods

Assume that as a result of a sales tax t_k imposed on good k , the price paid by consumers, p_k^C , increases by $p_k^C = (1 + t_j) p_k^P$, where p_k^P is the price received by producers. In this circumstance, if the tax on goods 1 and 2 coincides (i.e., $t_1 = t_2$), the price ratio consumers and producers face is unaffected because